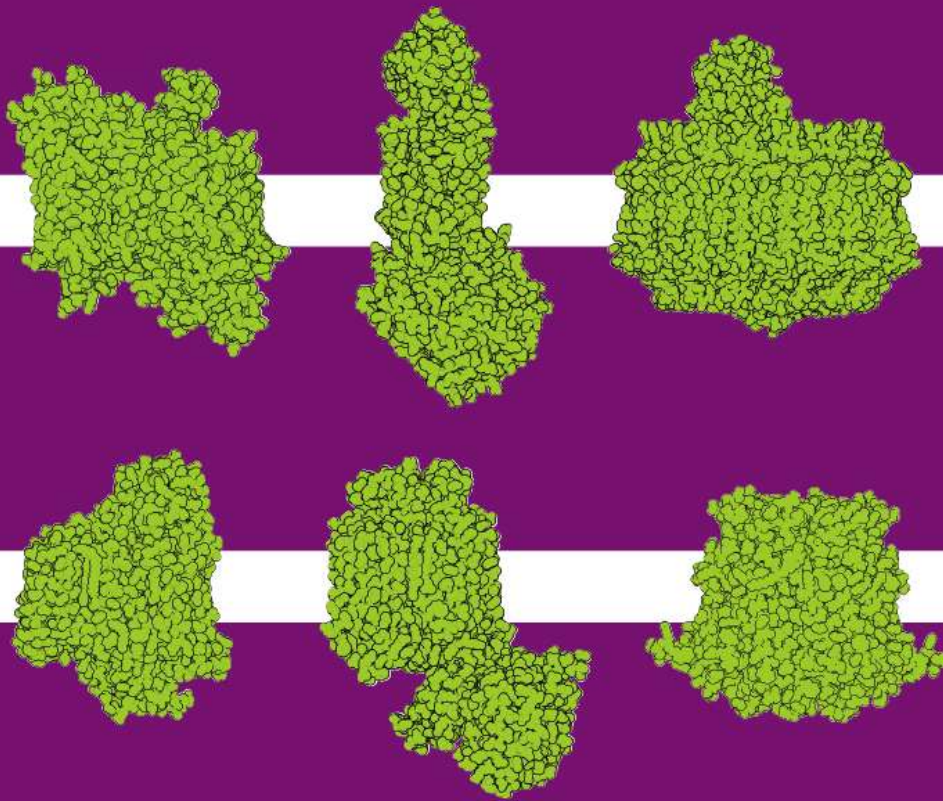


MOLECULAR BIOLOGY OF
THE CELL
SEVENTH EDITION



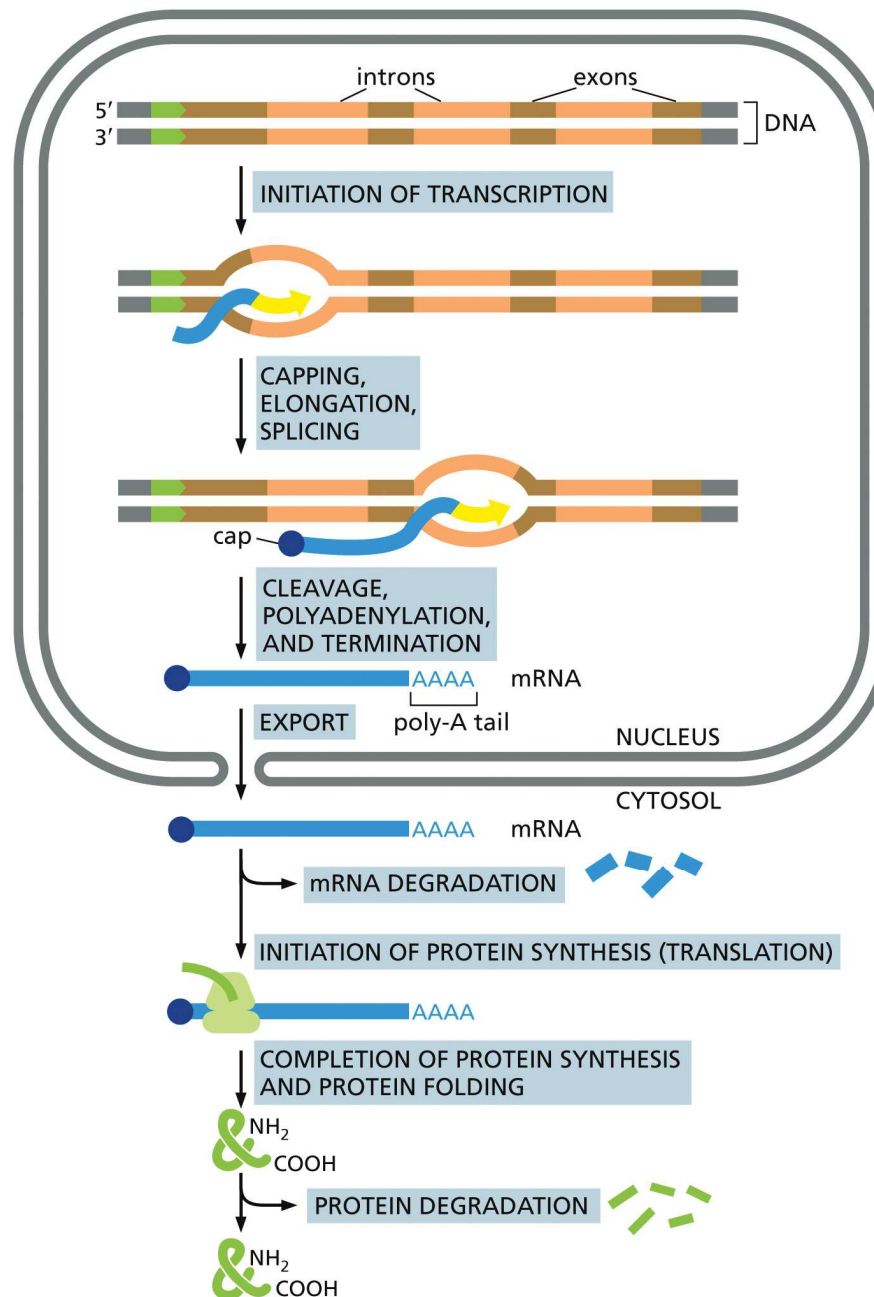
ALBERTS HEALD JOHNSON MORGAN RAFF ROBERTS WALTER

Chapter 6

**How Cells Read the Genome:
From DNA to Protein**

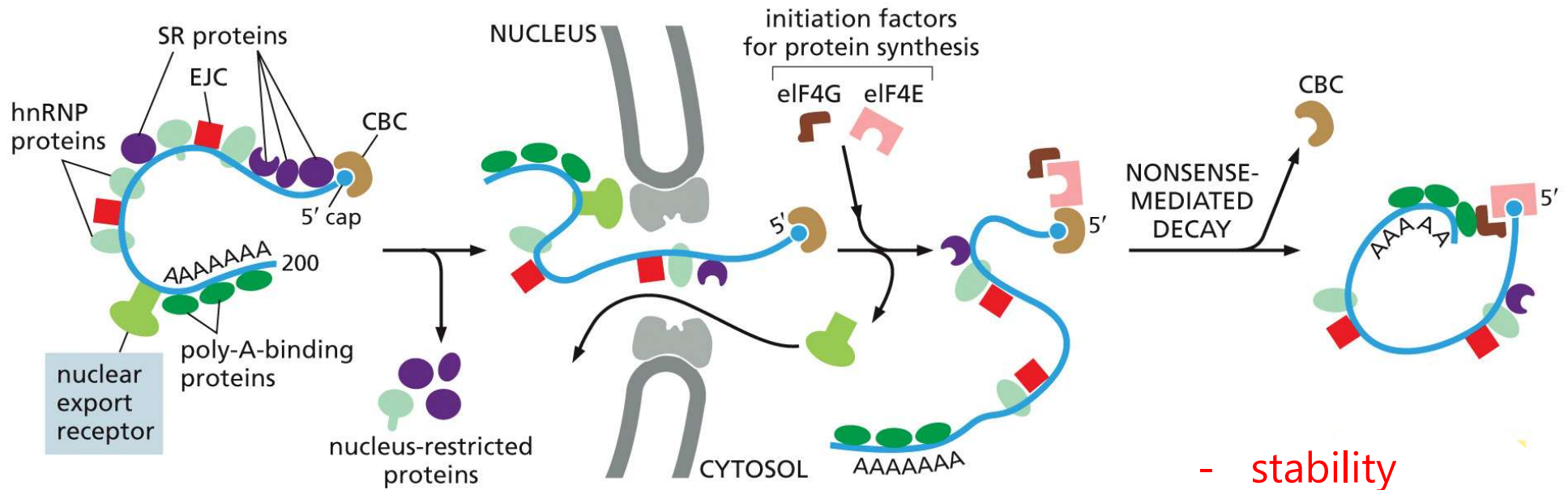
PART 2

From DNA to RNA to Protein to Degradation



Regulatory steps
in protein synthesis

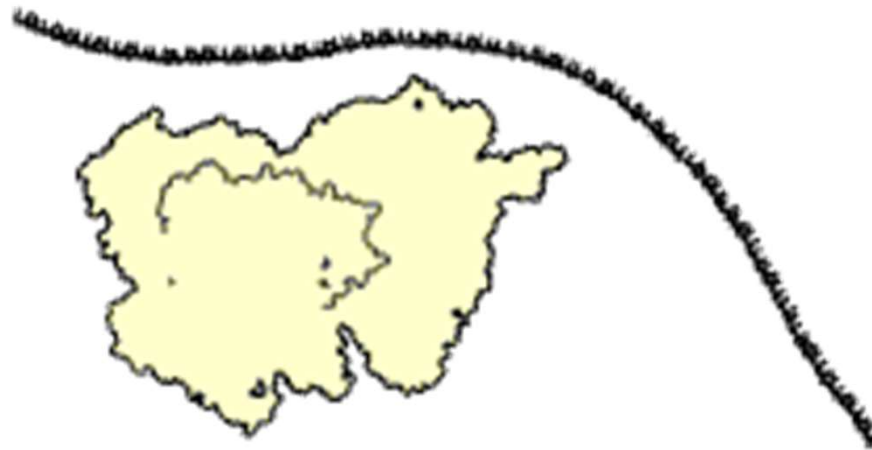
Export-ready mRNA and its transport



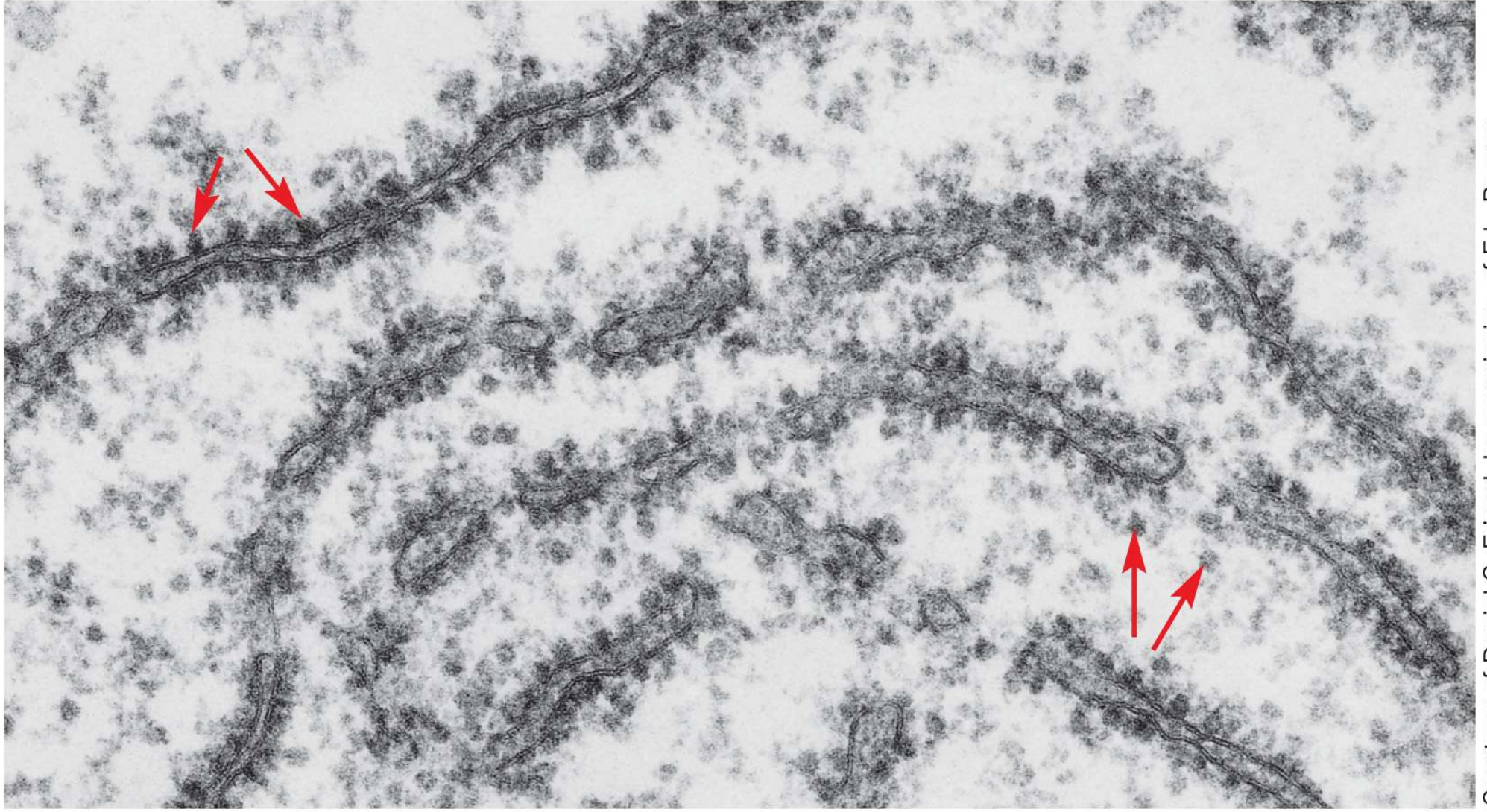
- stability
- quality control
- efficiency

And now...?

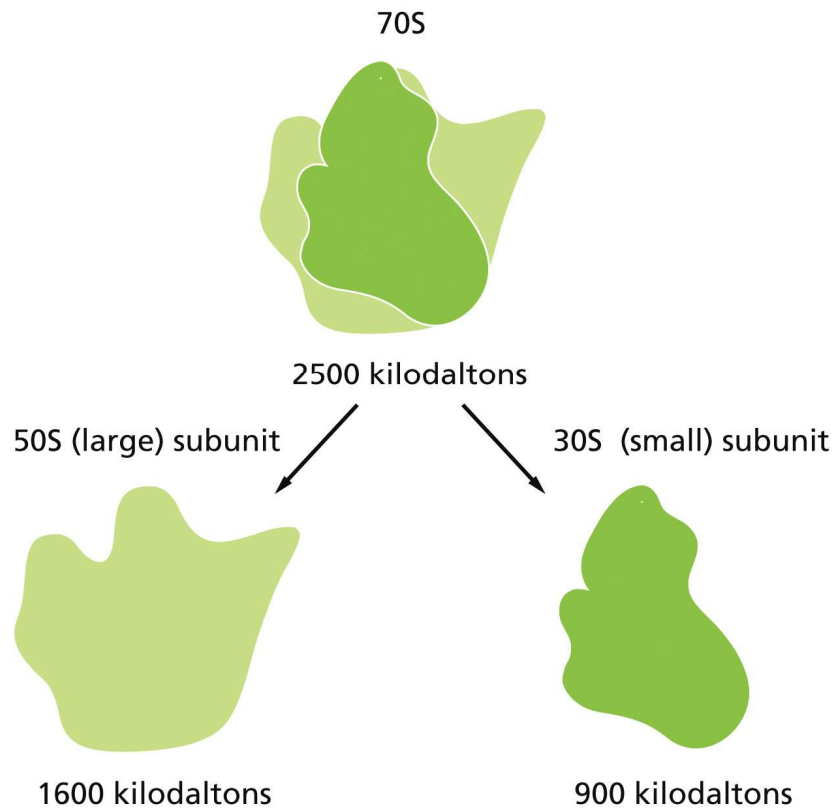
A ribosome translating a protein



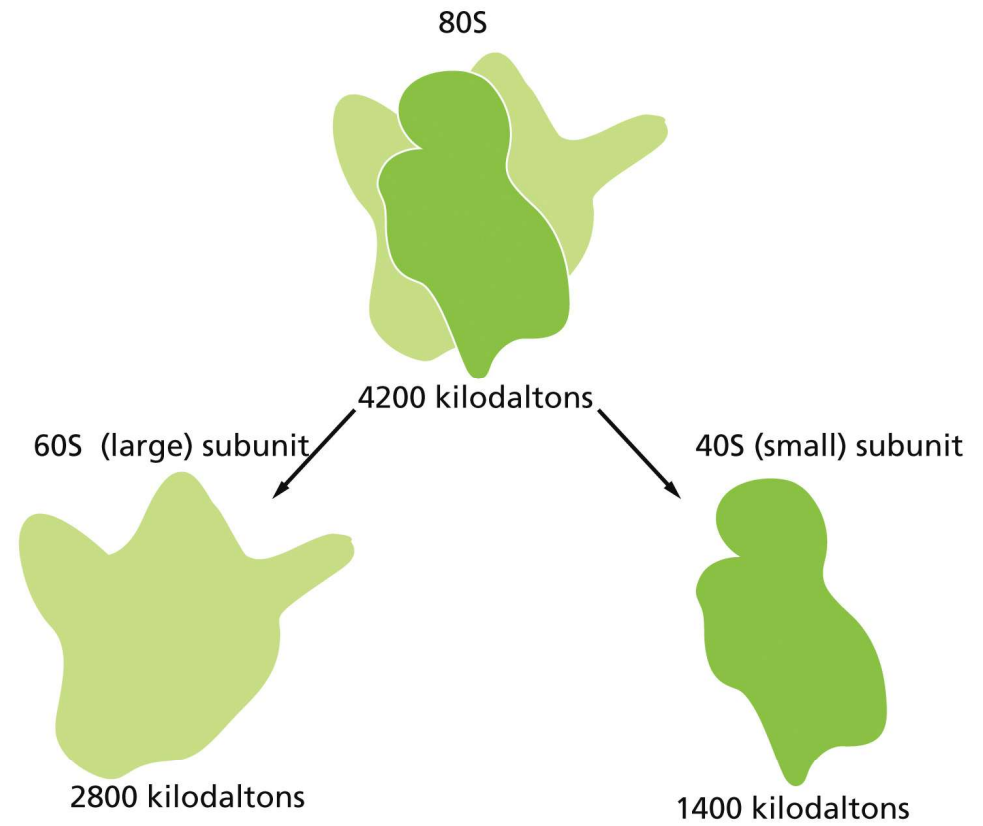
By Bensaccount at en.wikipedia, CC BY 3.0,
<https://commons.wikimedia.org/w/index.php?curid=8287100>



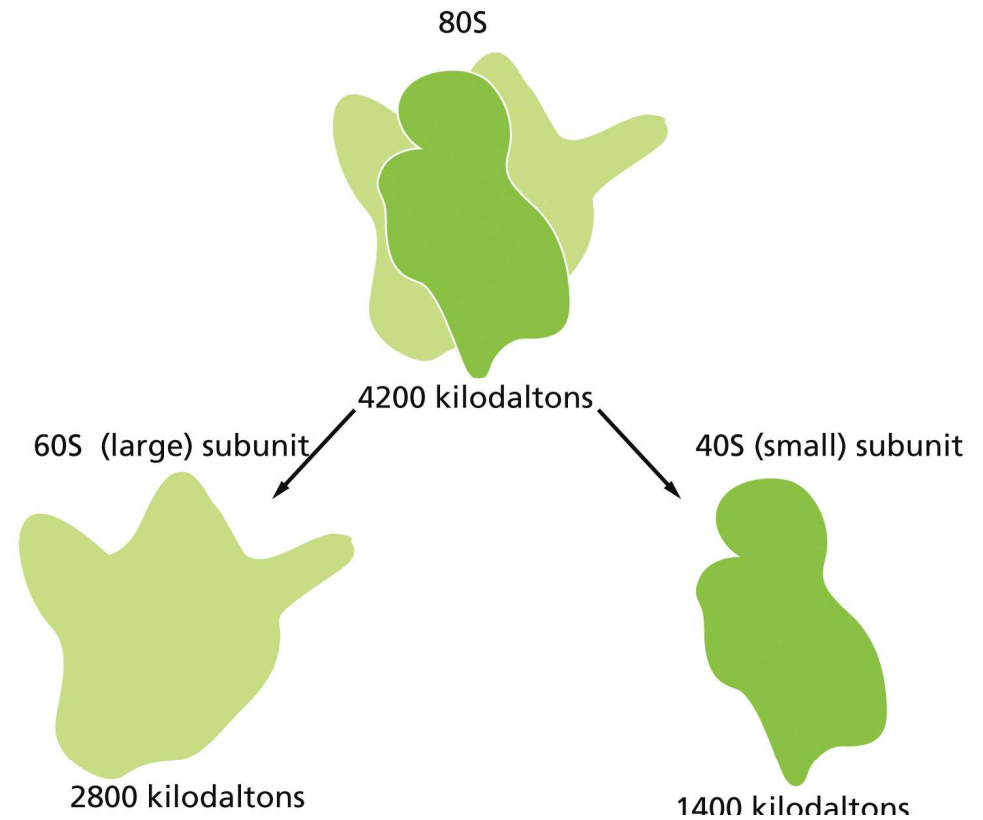
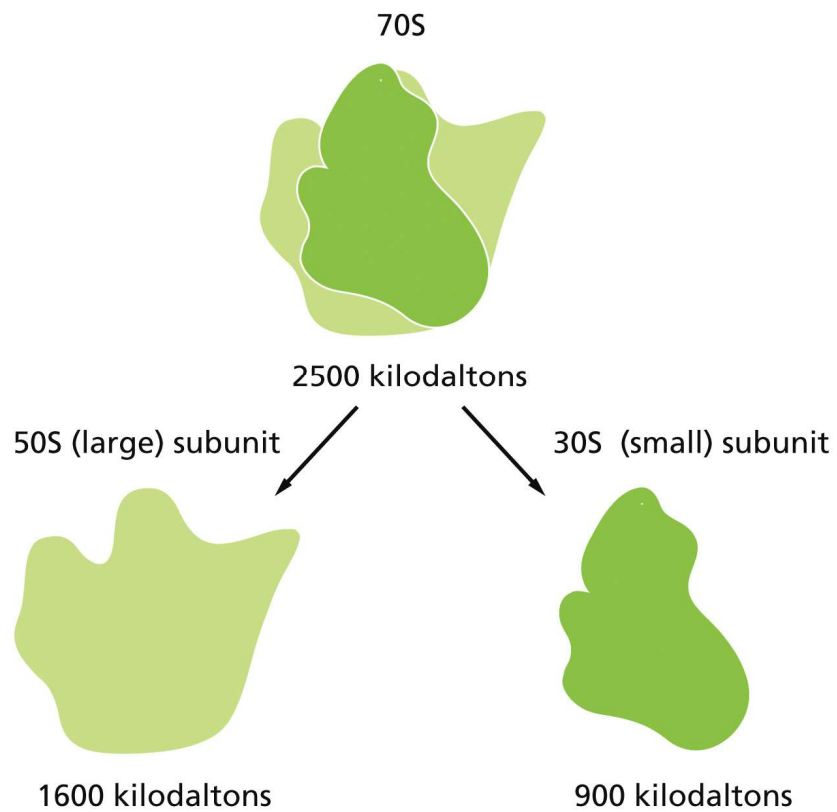
400 nm



BACTERIAL RIBOSOME



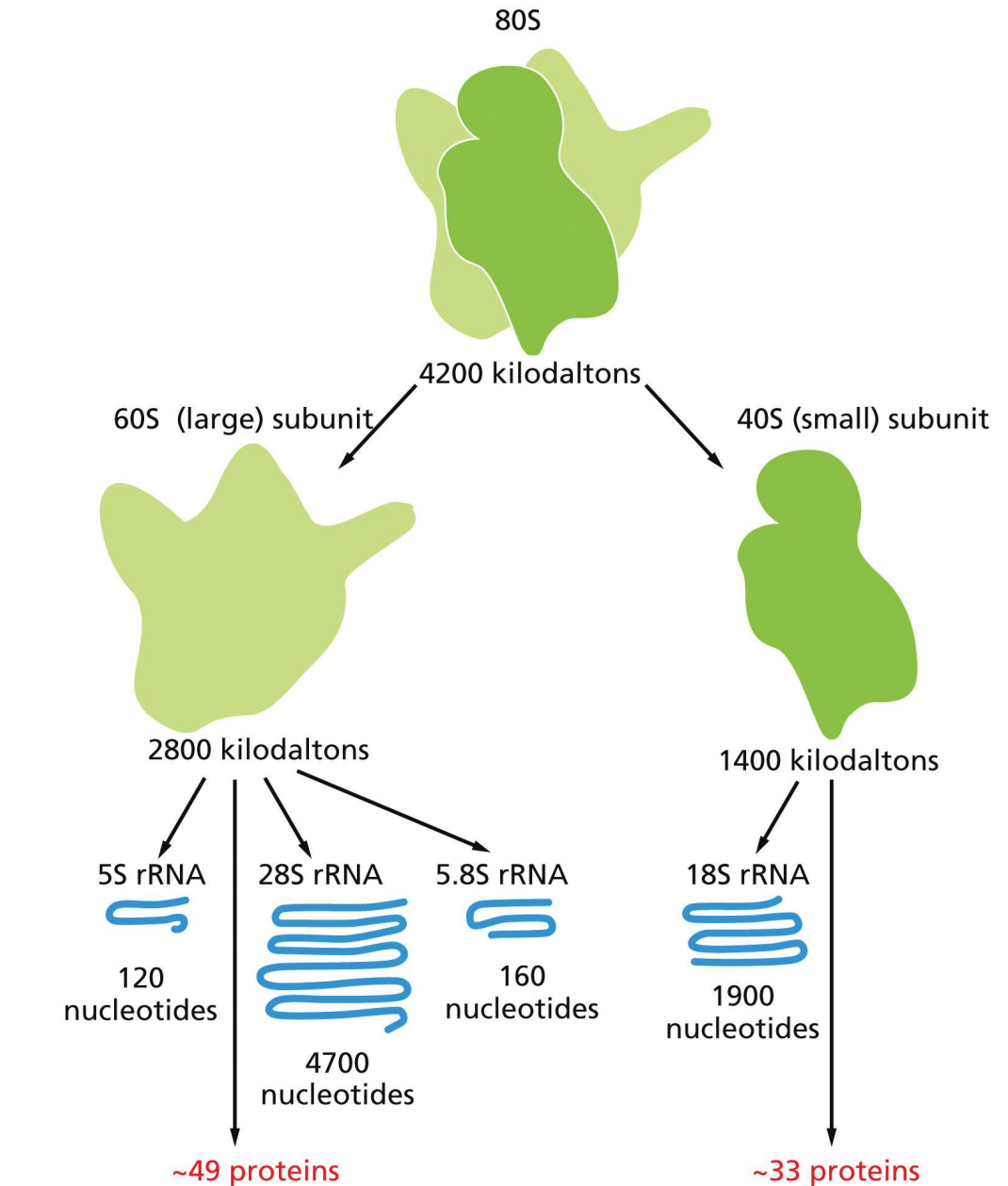
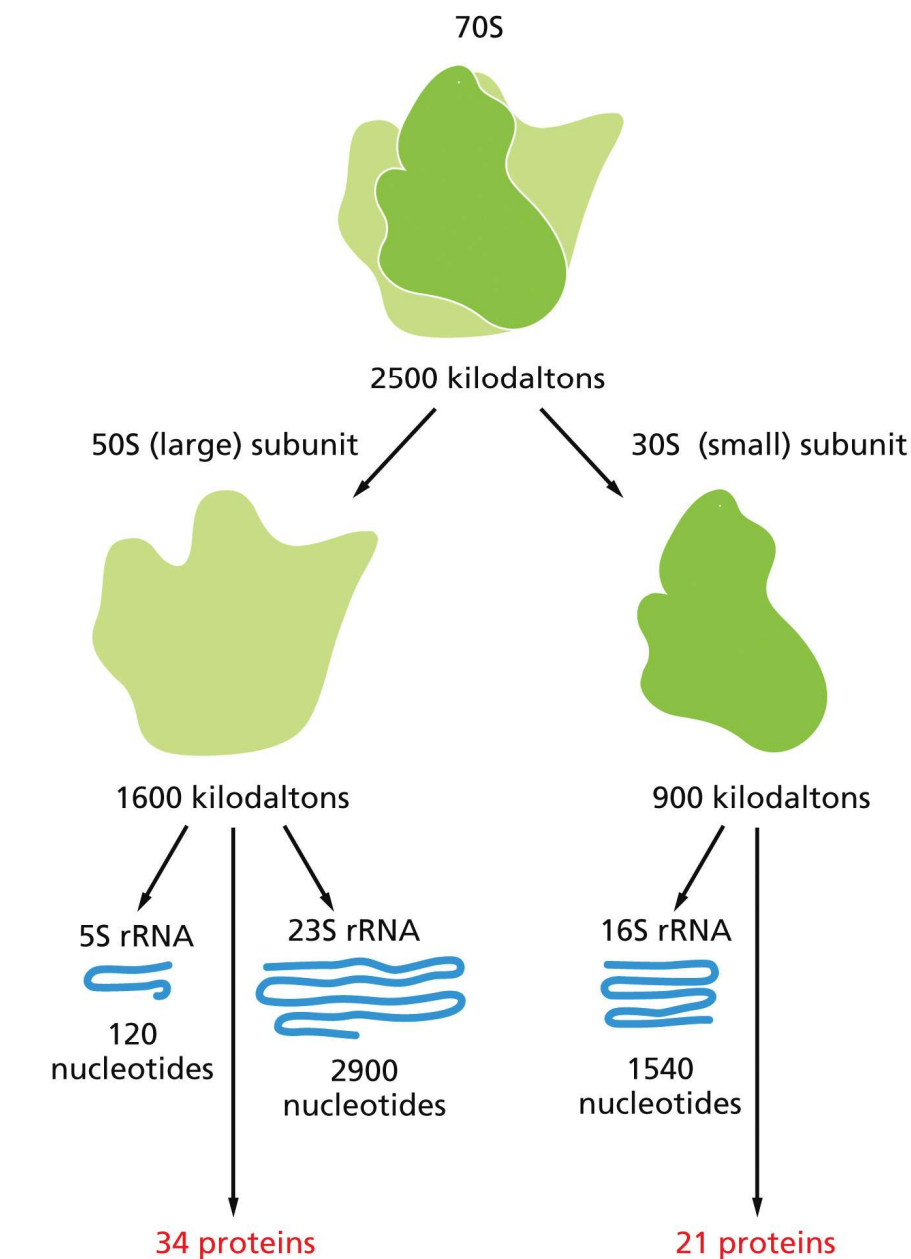
EUKARYOTIC RIBOSOME

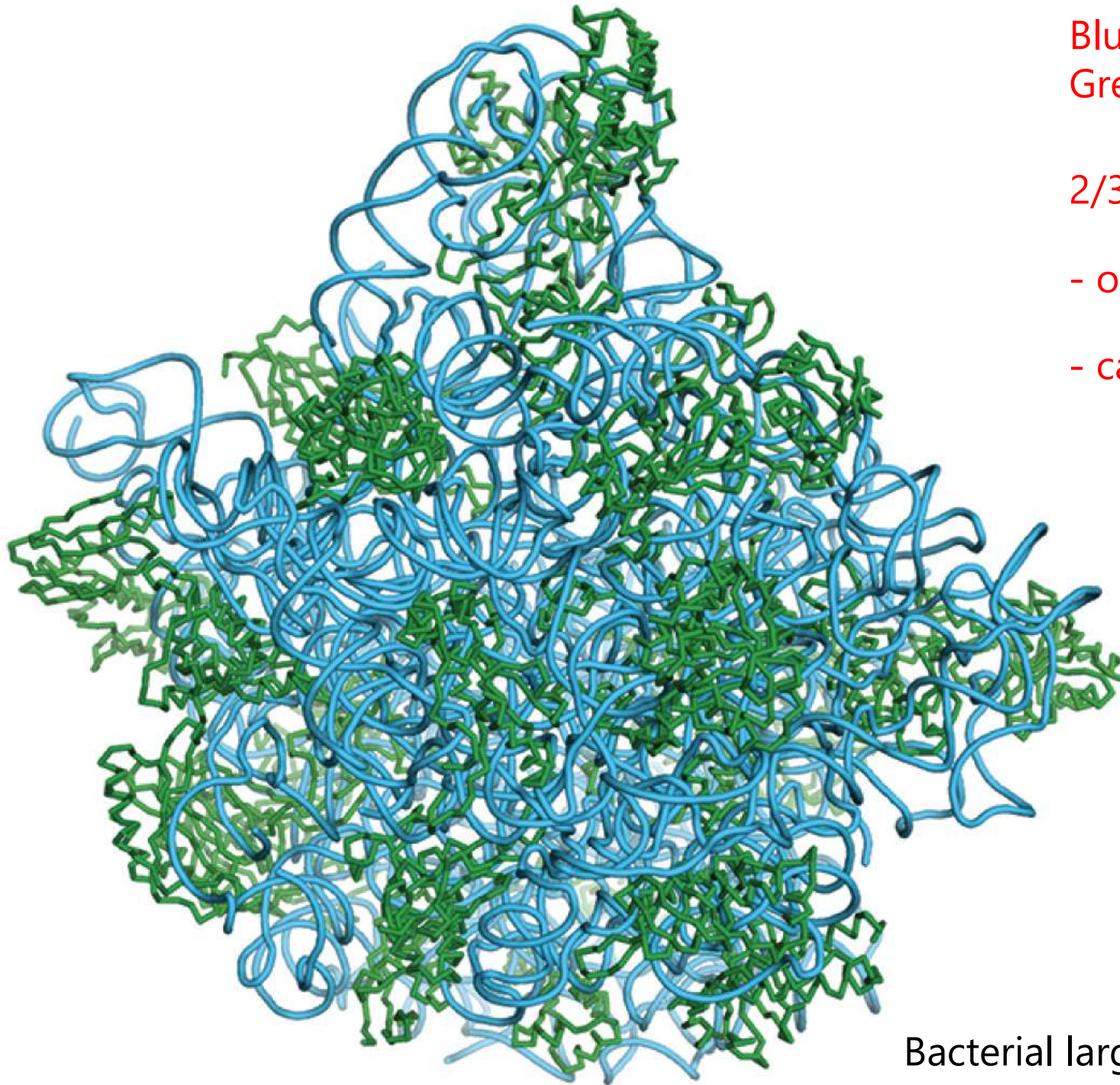


Building block(s) of ribosomes?

BACTERIAL RIBOSOME

EUKARYOTIC RIBOSOME





Blue?
Green?

2/3 RNA & 1/3 protein

- overall structure

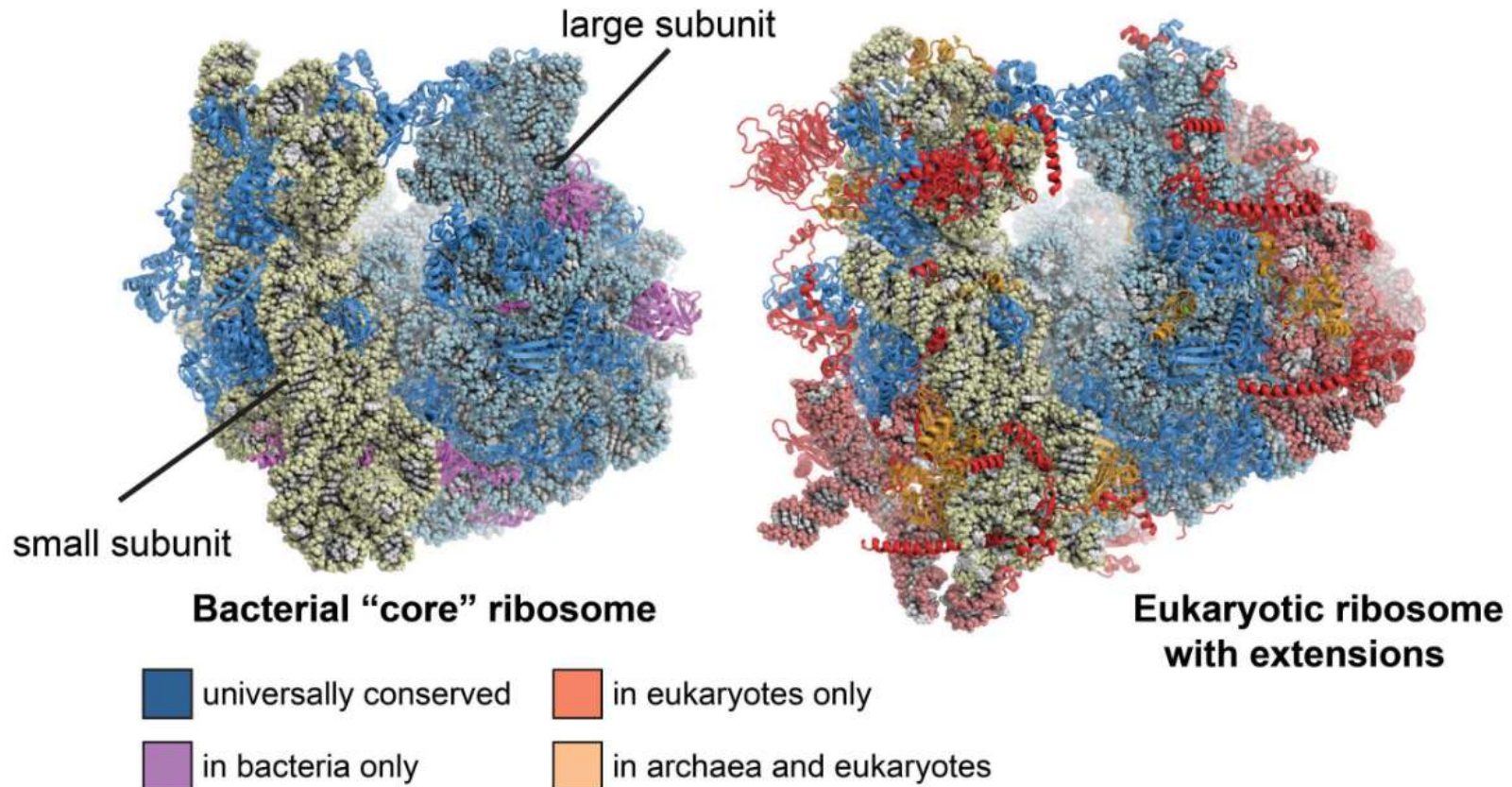
- catalytic activity

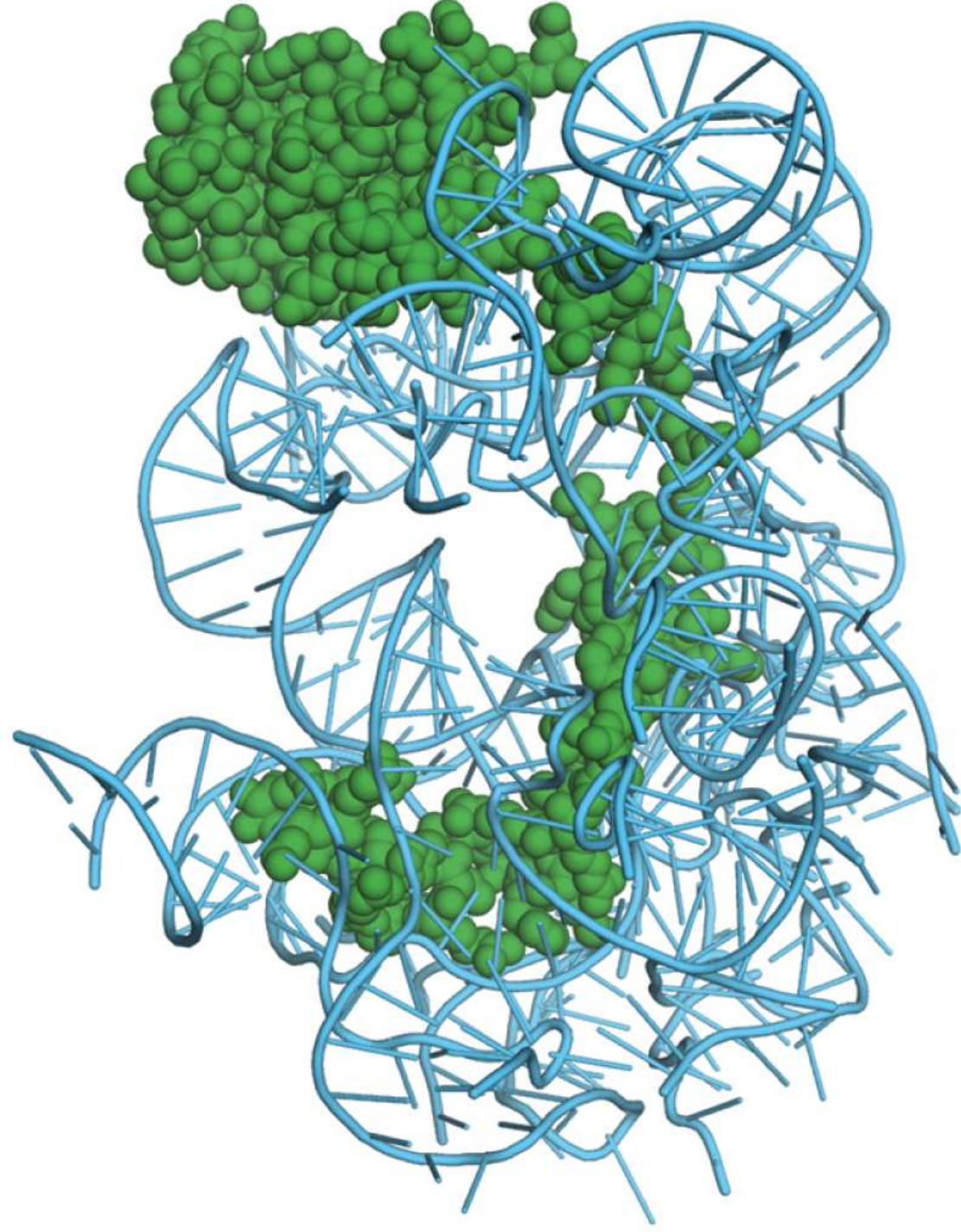
Bacterial large ribosomal subunit
(5S and 23S)

From first lecture: rRNA as example of sequence conservation

```

GTTCCGGGGGGAGTATGGTTGCAAAGCTGAAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC human
GCCGCCTGGGGAGTACGGTCGCAAGACTGAAACTTAAAGGAATTGGCGGGGAGCACTACAACGGGTGGAGCCTGCGGTTTAATTGGATTCAACGCCGGGCATCTTACCA Methanococcus
ACCGCCTGGGGAGTACGGCCGCAAGGTTAAACTCAAATGAATTGACGGGGGCCCGC•ACAAGCGGTGGAGCATGTGGTTTAATTTCGATGCAACGCGAAGAACCTTACCT E. coli
GTTCCGGGGGGAGTATGGTTGCAAAGCTGAAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC human
  
```





From D. Klein et al., *J. Mol. Biol.* 340:141–177, 2004. PDB code: 1S72.

ribosomes per cell: 10^6 - 10^7

rRNA gene copies per haploid genome: ~200

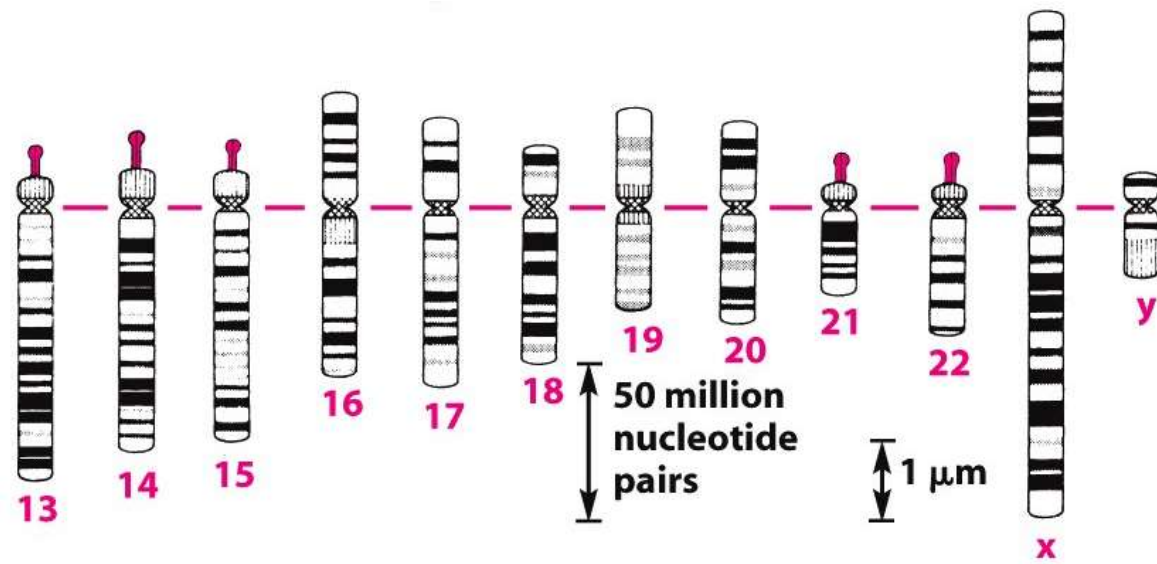
ribosomes per cell: 10^6 - 10^7

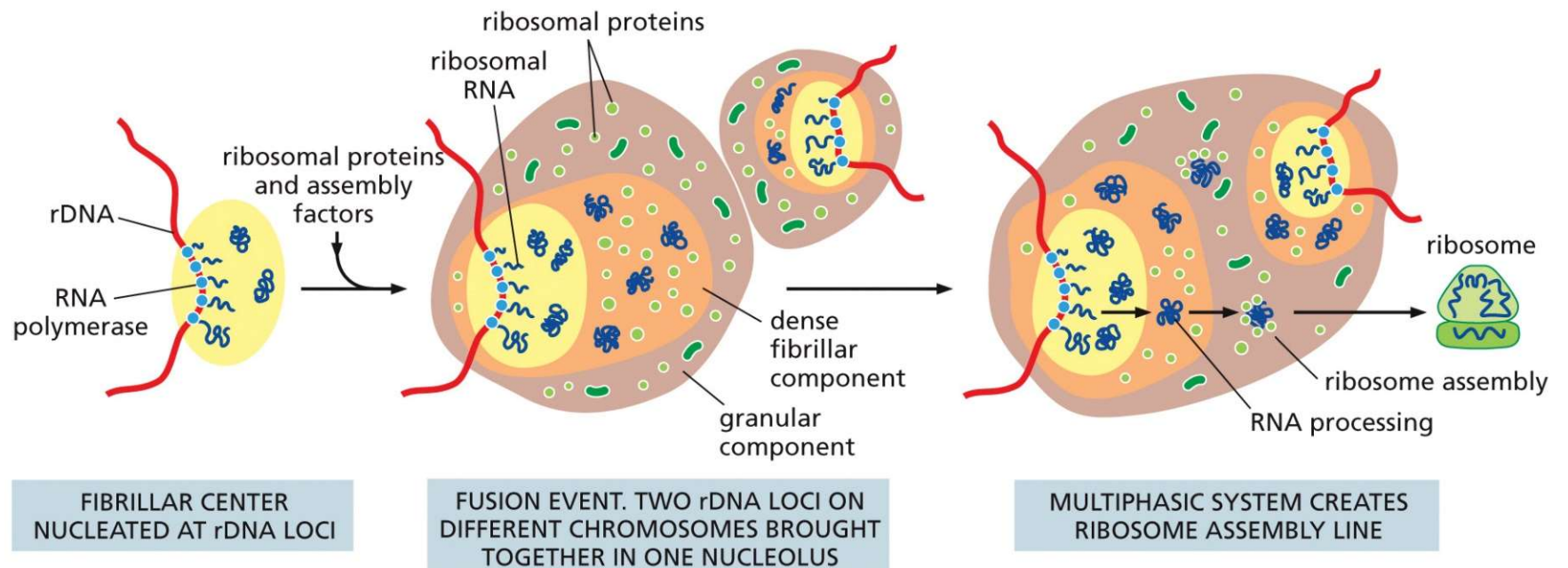
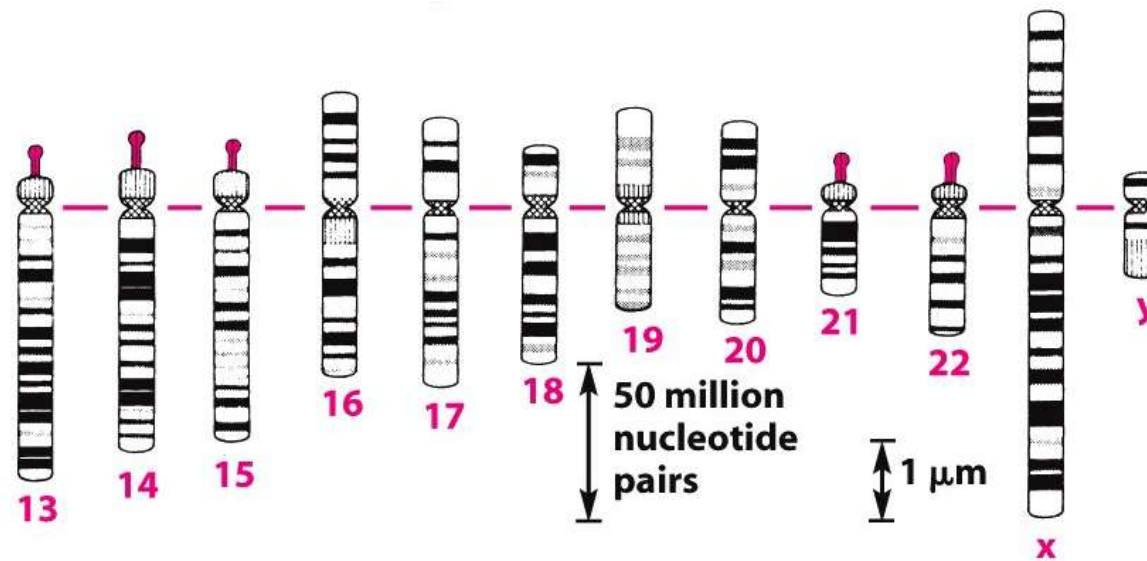
rRNA gene copies per haploid genome: ~200

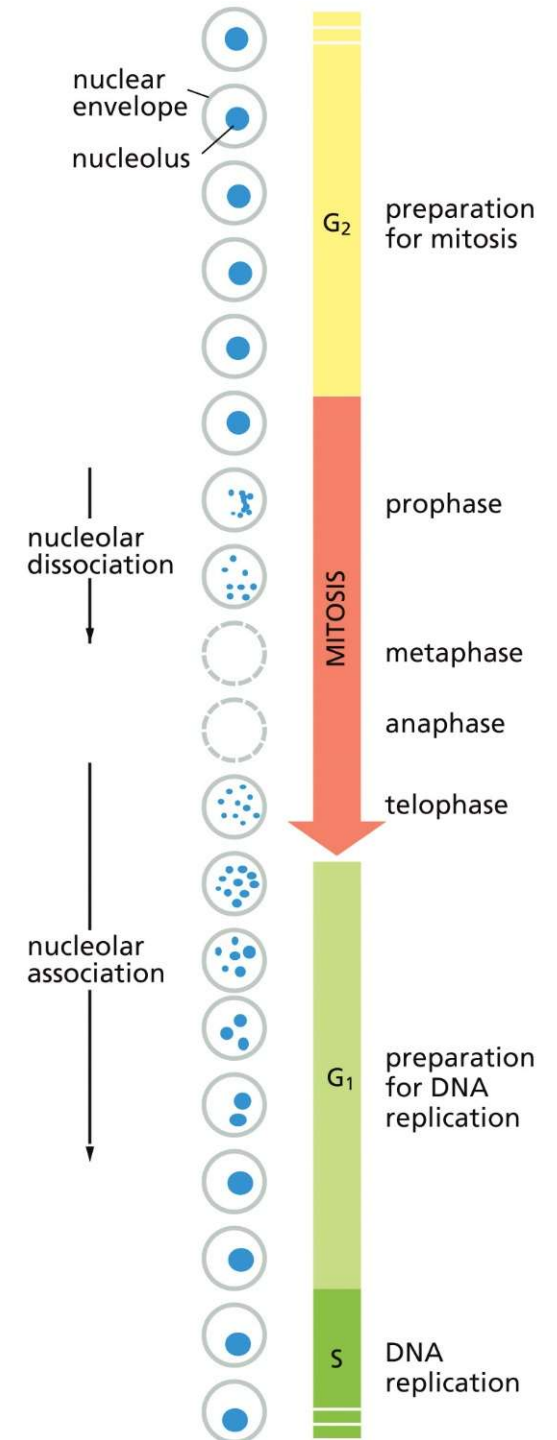
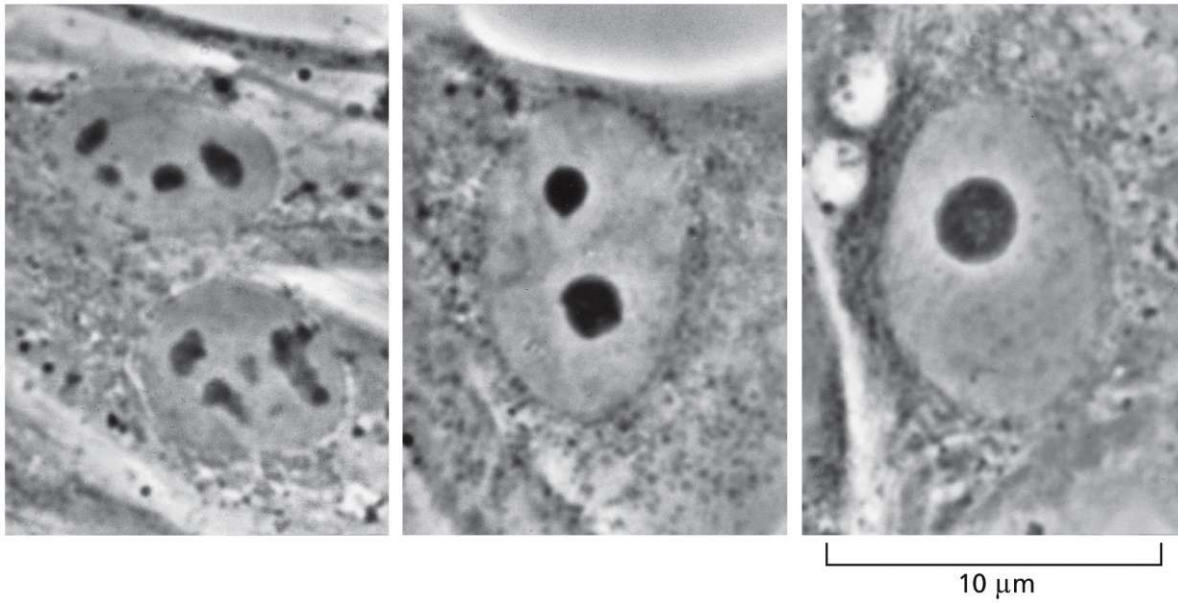


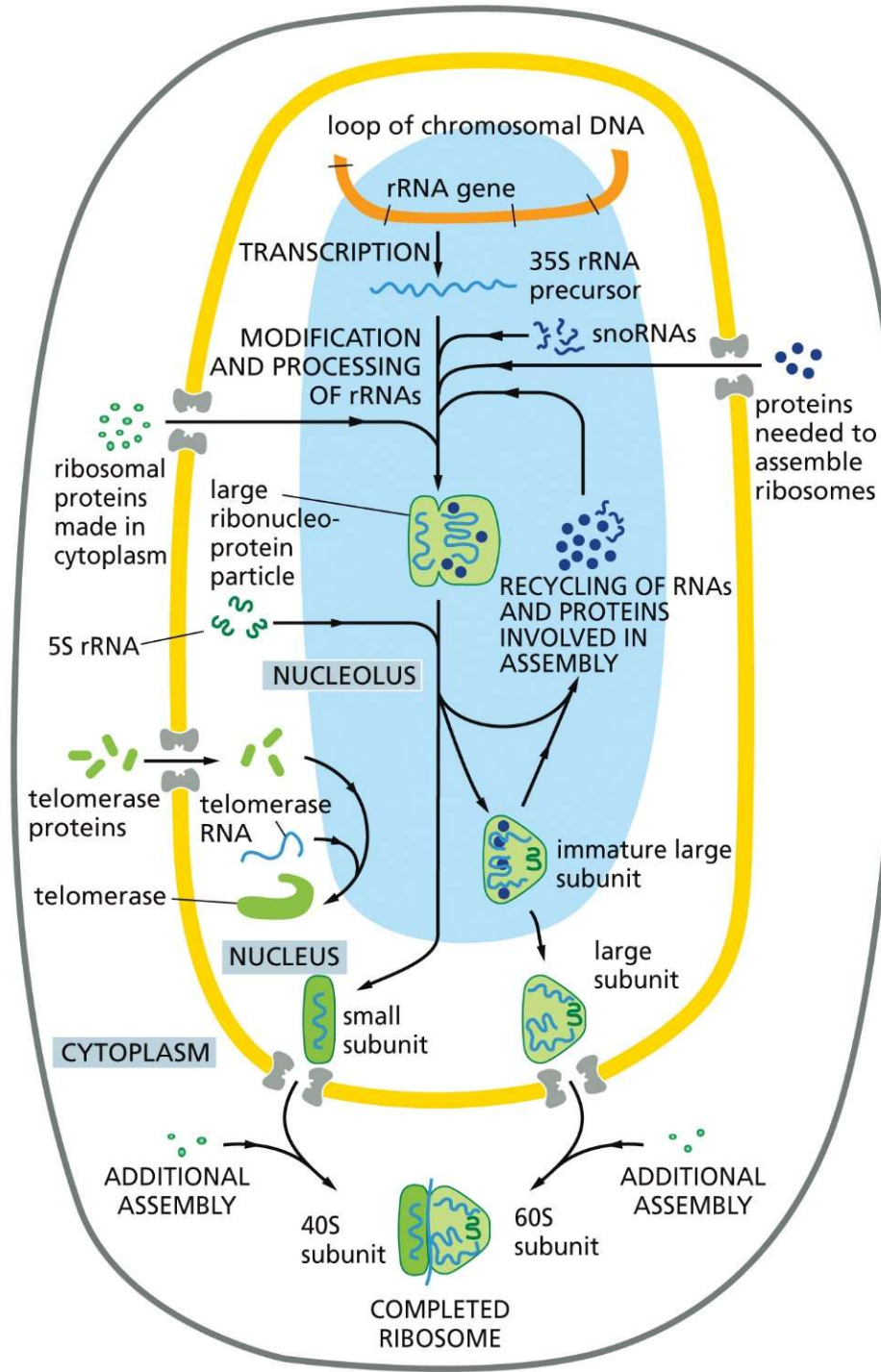
From V.E. Foe, *Cold Spring Harb. Symp. Quant. Biol.* 42:723–740, 1978. With permission from Cold Spring Harbor Laboratory Press.

2 μm

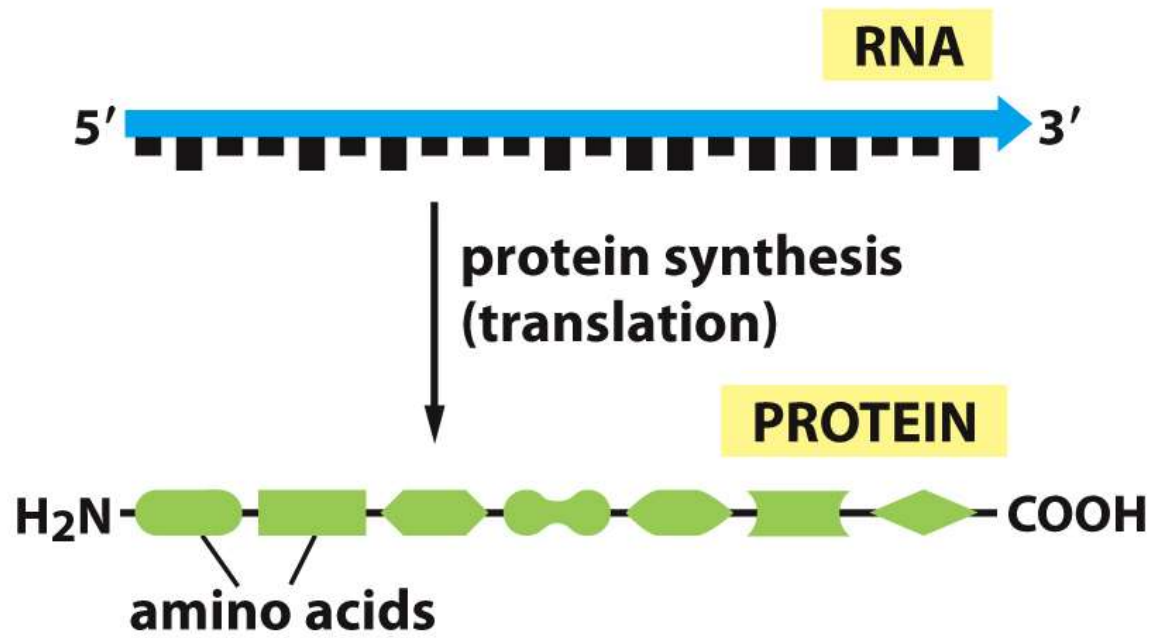








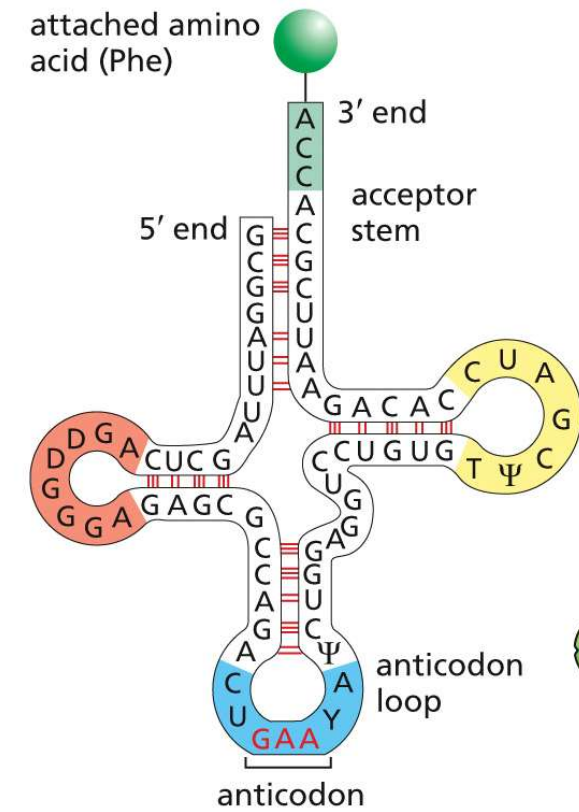
Adapted from A.R. Strom and C.P. Brangwynne, *J. Cell Sci.* 132:jcs235093, 2019. With permission from the Company of Biologists.



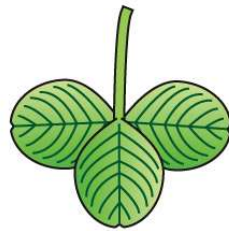
+ ribosomes

+ ?

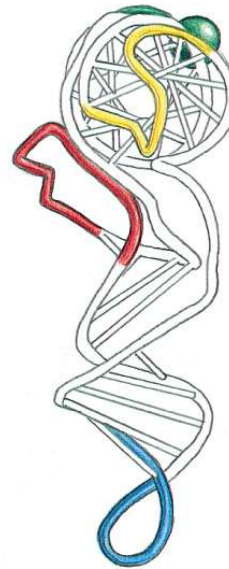
transfer RNA (tRNA)



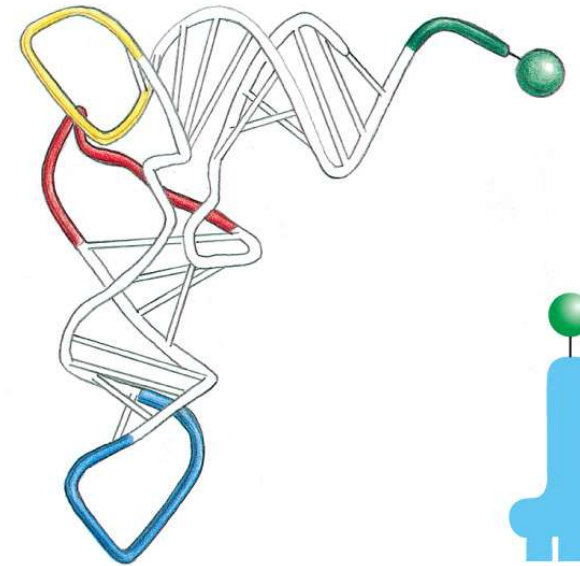
(A)



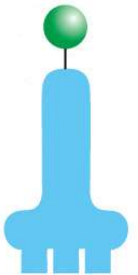
a cloverleaf



(B)



(C)



(D)

5' GCGGAUUUAGCUCAGDDGGGAGAGCGCCAGACUGAAYAYΨCUGGAGGUCCUGUGTΨCGAUCCACAGAAUUCGCACCA 3'

anticodon

(E)

The universal genetic code

4 different nucleotides

3 consecutive nucleotides: codon

How many possible combinations?

$$4 \times 4 \times 4 = 64$$

How many different amino acids commonly found in proteins?

So: some triplets never used, or redundancy?

The universal genetic code

GCA	AGA									UUA					AGC					
GCC	AGG									UUG					AGU					
GCG	CGA						GGA		AUA	CUA				CCA	UCA	ACA			GUA	
GCU	CGC						GGC		AUC	CUC				CCC	UCC	ACC			GUC	UAA
	CGG	GAC	AAC	UGC	GAA	CAA	GGG	CAC	AUU	CUG	AAA			UUC	CCG	UCG	ACG		UAG	
	CGU	GAU	AAU	UGU	GAG	CAG	GGU	CAU	AUU	CUU	AAG	AUG	UUU	CCU	UCU	ACU	UGG	UAU	GUG	UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

TABLE 14-3 Some Differences Between the “Universal” Code and Mitochondrial Genetic Codes*

Codon	“Universal” code	Mitochondrial codes			
		Mammals	Invertebrates	Yeasts	Plants
UGA	STOP	<i>Trp</i>	<i>Trp</i>	<i>Trp</i>	STOP
AUA	Ile	<i>Met</i>	<i>Met</i>	<i>Met</i>	Ile
CUA	Leu	Leu	Leu	<i>Thr</i>	Leu
AGA AGG	Arg	<i>STOP</i>	<i>Ser</i>	Arg	Arg
*Red italics indicate that the code differs from the “Universal” code.					

Table 14-4

The universal genetic code

GCA	AGA									UUA					AGC					
GCC	AGG									UUG					AGU					
GCG	CGA						GGA			CUA				CCA	UCA	ACA			GUA	
GCU	CGC						GGC		AUA	CUC				CCC	UCC	ACC			GUC	UAA
	CGG	GAC	AAC	UGC	GAA	CAA	GGG	CAC	AUC	CUG	AAA		UUC	CCG	UCG	ACG		UAC	GUG	UAG
	CGU	GAU	AAU	UGU	GAG	CAG	GGU	CAU	AUU	CUU	AAG	AUG	UUU	CCU	UCU	ACU	UGG	UAU	GUU	UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

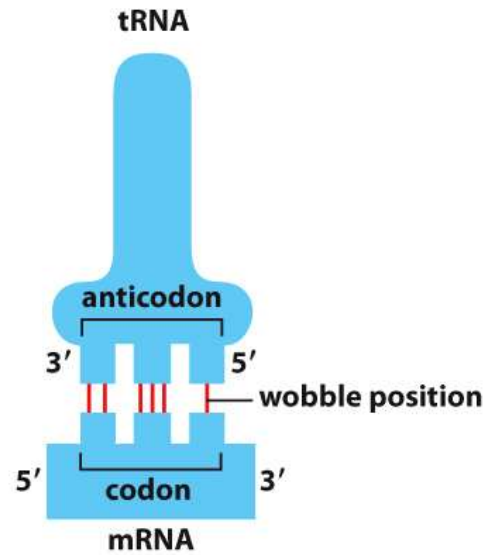
Redundancy

Mechanism(s)?

- More than 1 tRNA for many of the aa's
- Some tRNA's can base-pair with more than 1 codon

Bacteria: 31 kinds of tRNA molecules

Humans: ~500 tRNA genes, representing 48 tRNA's



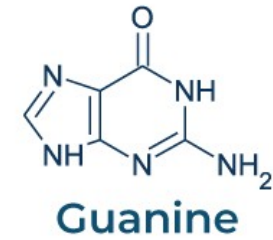
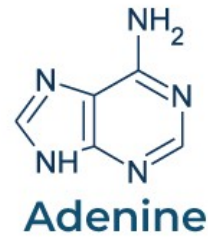
bacteria

wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U or I
G	C or U

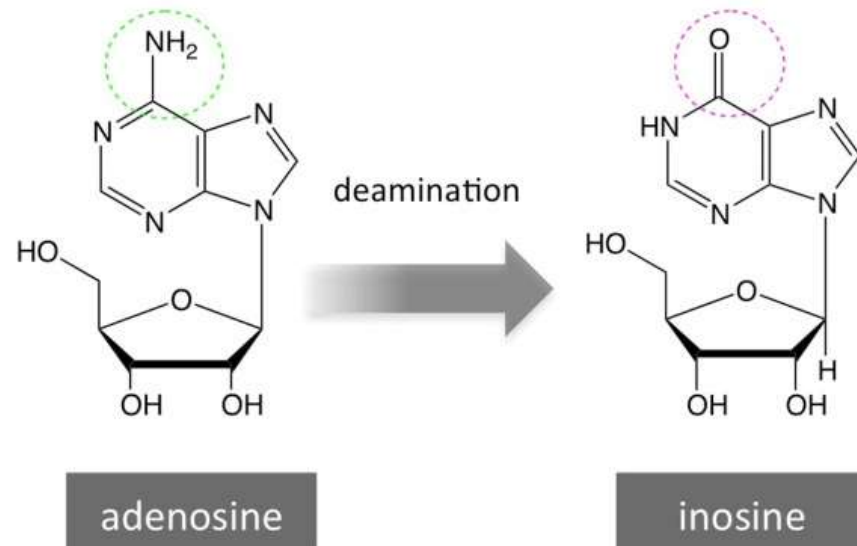
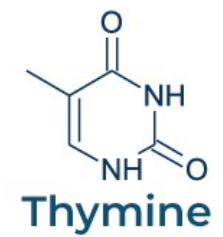
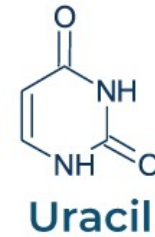
eukaryotes

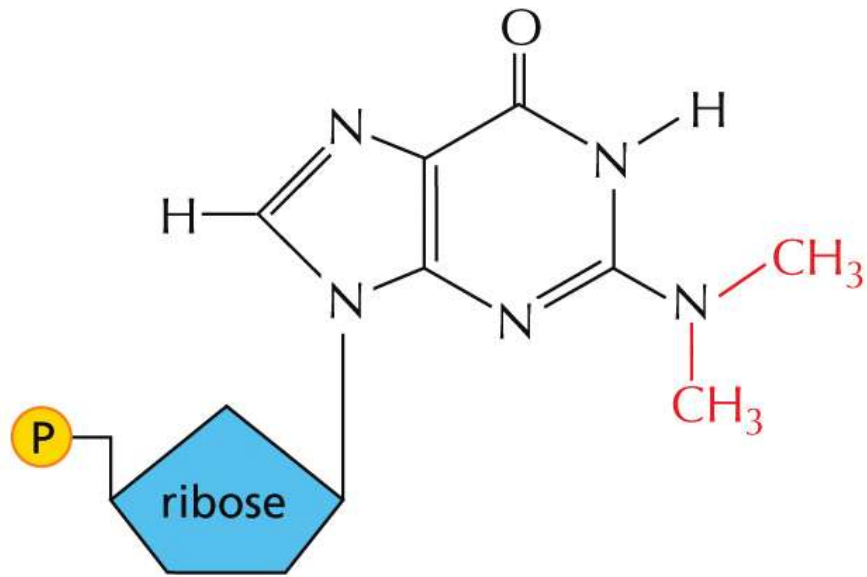
wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U
G	C

Purines

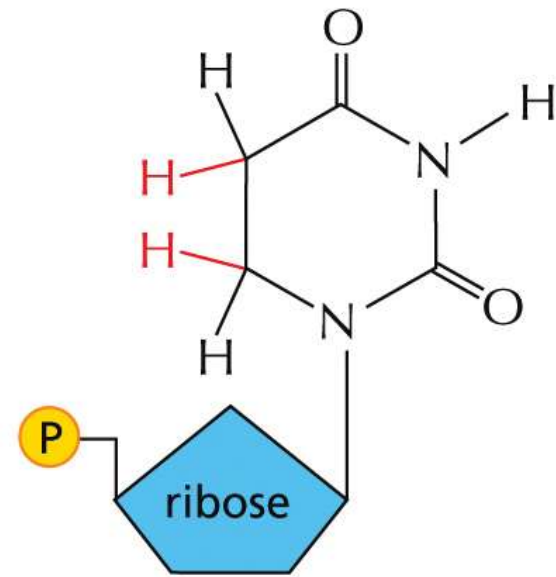


Pyrimidines

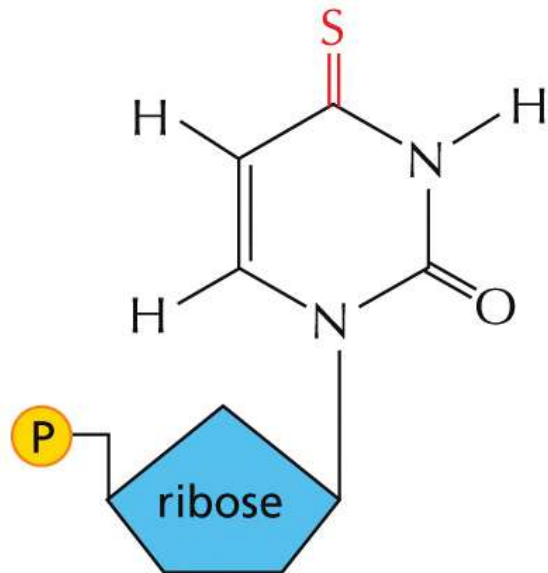




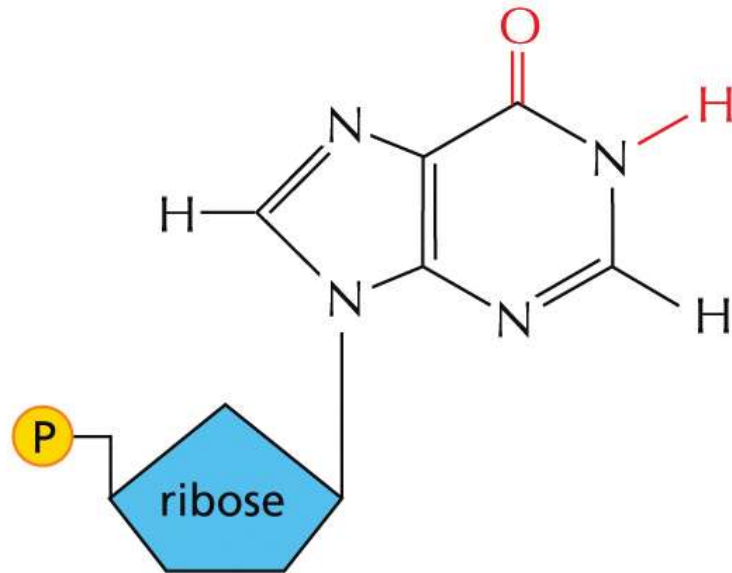
two methyl groups added to G
(*N,N*-dimethyl G)



two hydrogens added to U
(dihydro U)

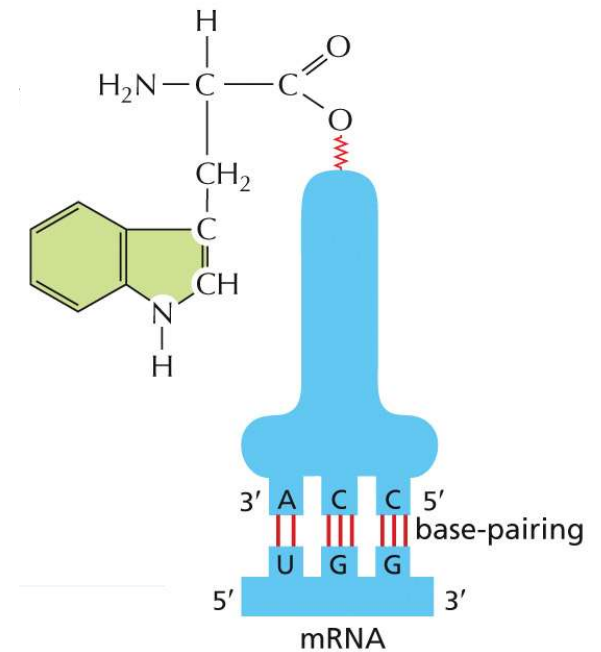


sulfur replaces oxygen in U
(4-thiouridine)



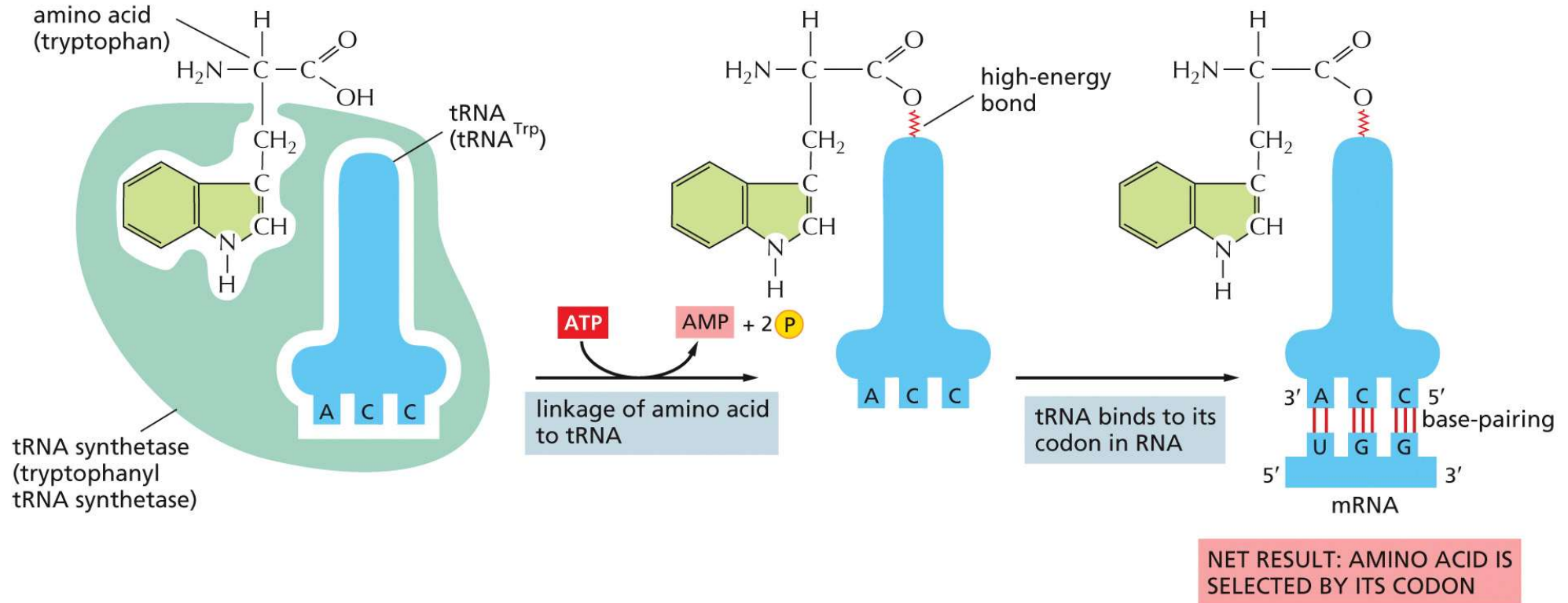
deamination of A
(inosine)

The genetic code is translated by means of **two** adaptors



NET RESULT: AMINO ACID IS
SELECTED BY ITS CODON

The genetic code is translated by means of **two** adaptors



A different synthetase enzyme for each aa or codon?

The recognition of a tRNA by its aminoacyl-tRNA synthetase

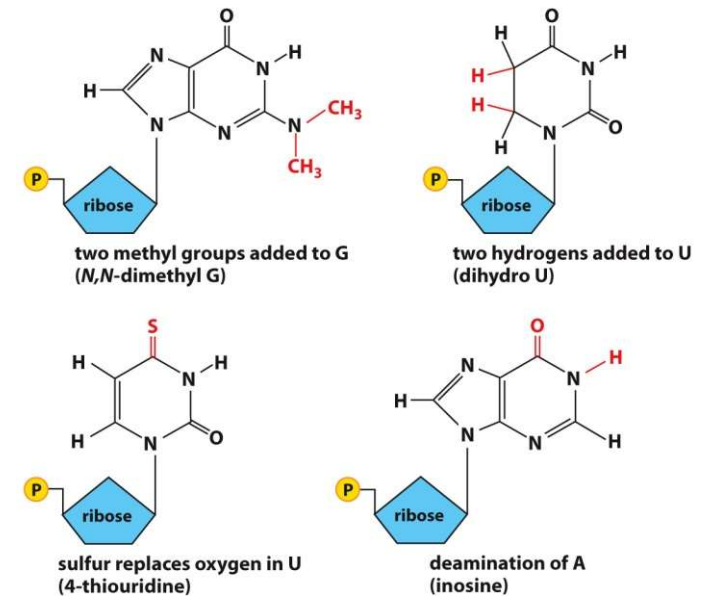
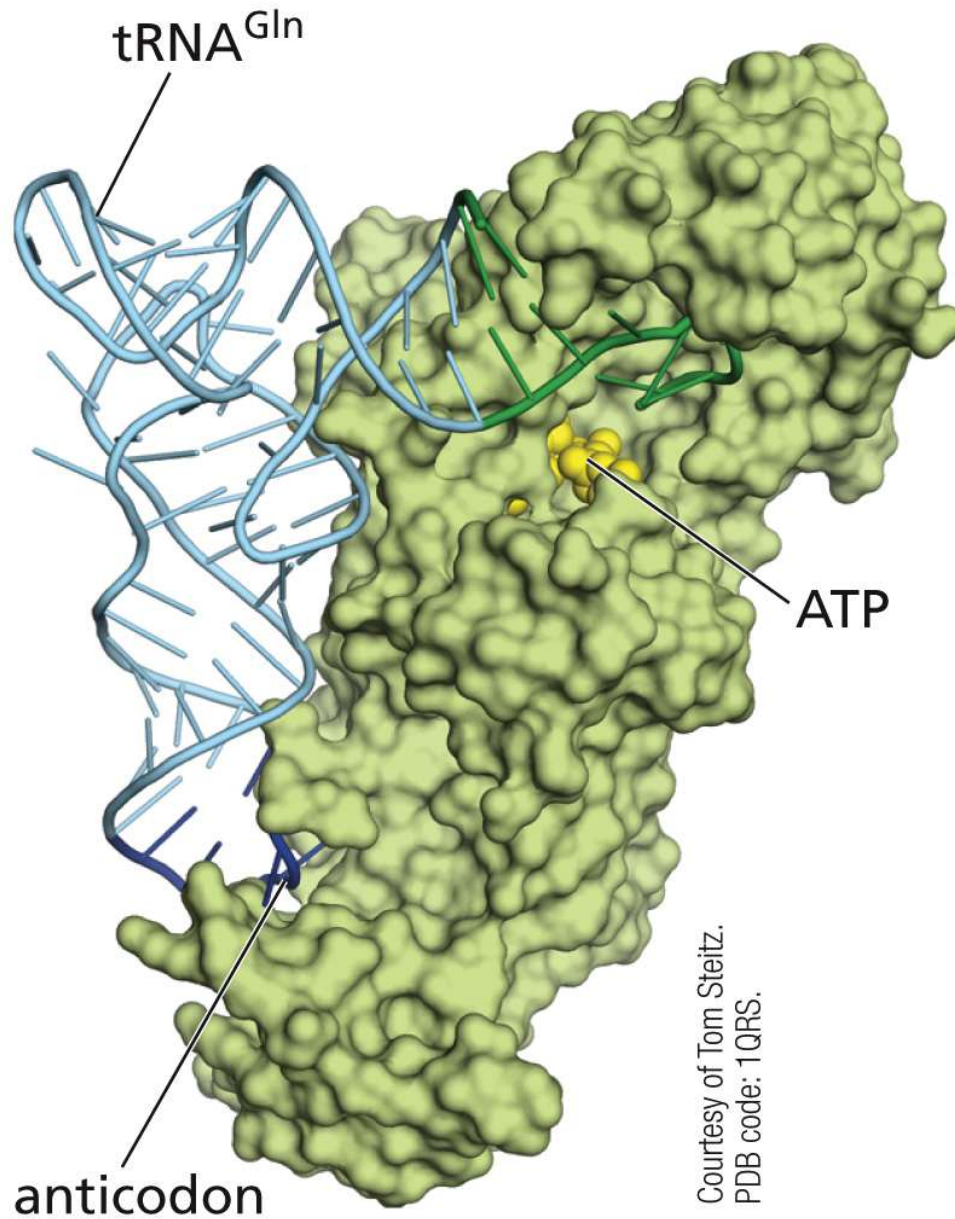
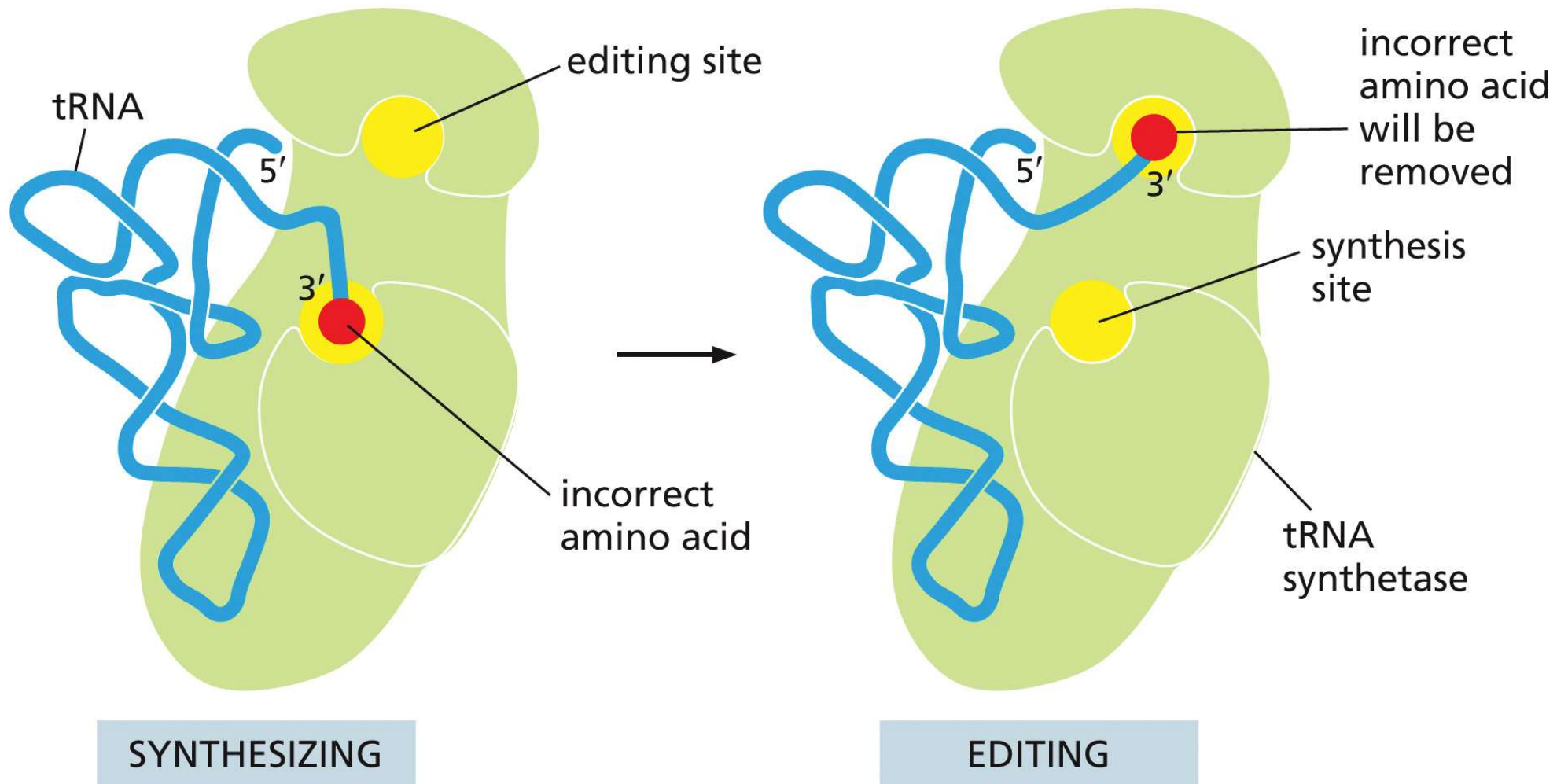
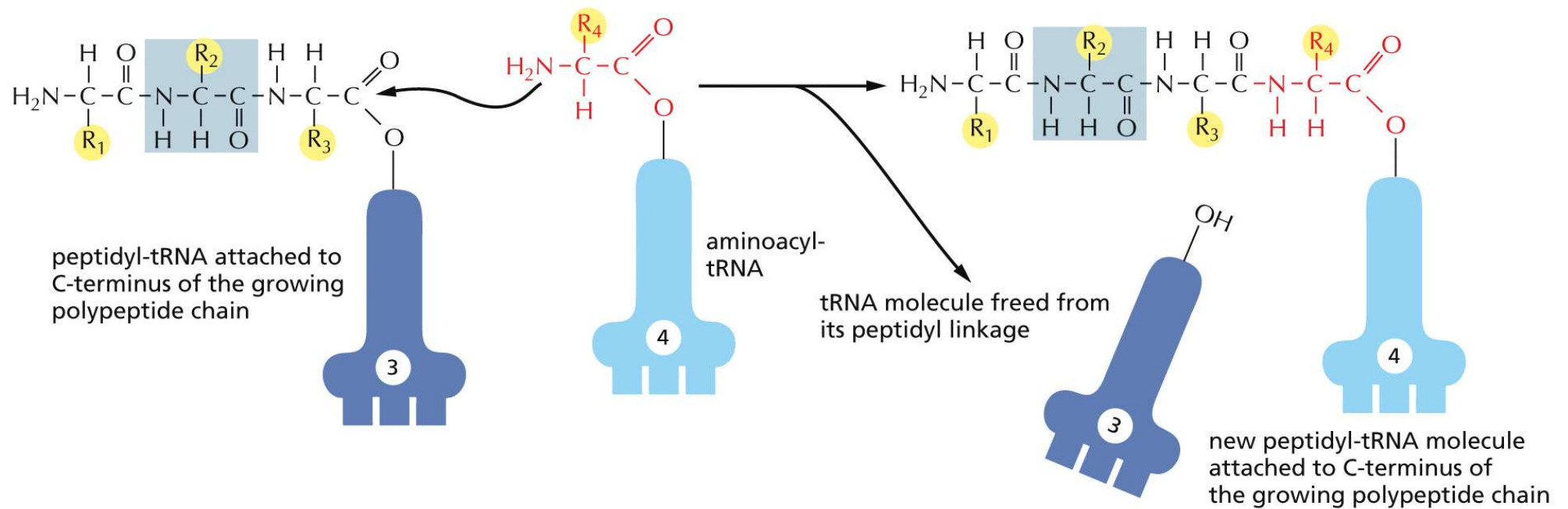


Figure 6-53 Molecular Biology of the Cell 6e (© Garland Science 2015)

Editing





POLYMER-END ACTIVATION (e.g., PROTEINS, FATTY ACIDS)

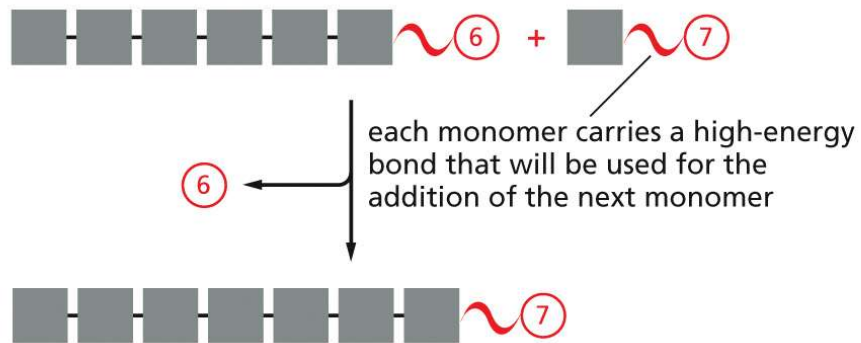
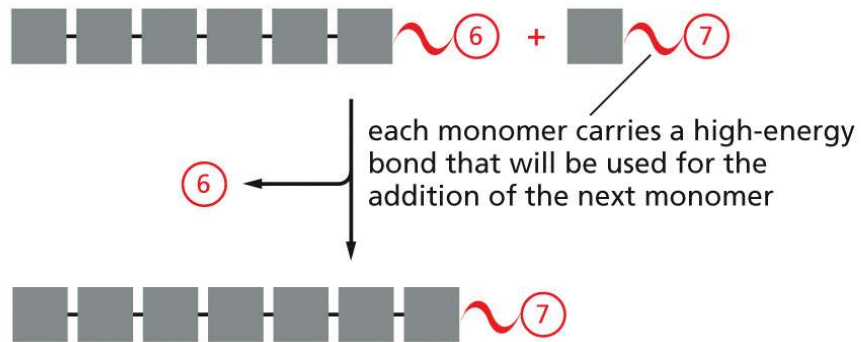


Figure 2-44

POLYMER-END ACTIVATION (e.g., PROTEINS, FATTY ACIDS)



DIRECT-MONOMER ACTIVATION (e.g., DNA, RNA, POLYSACCHARIDES)

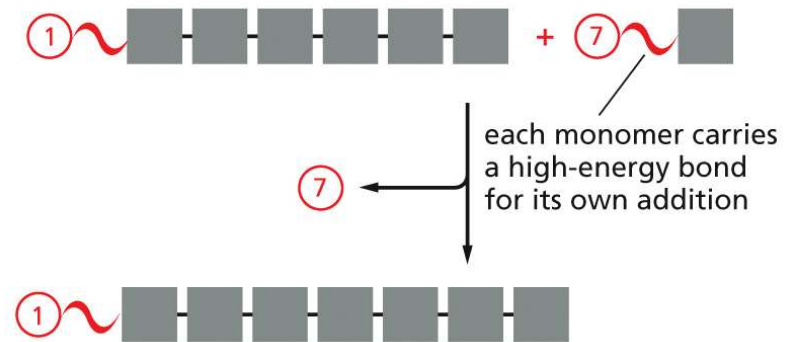
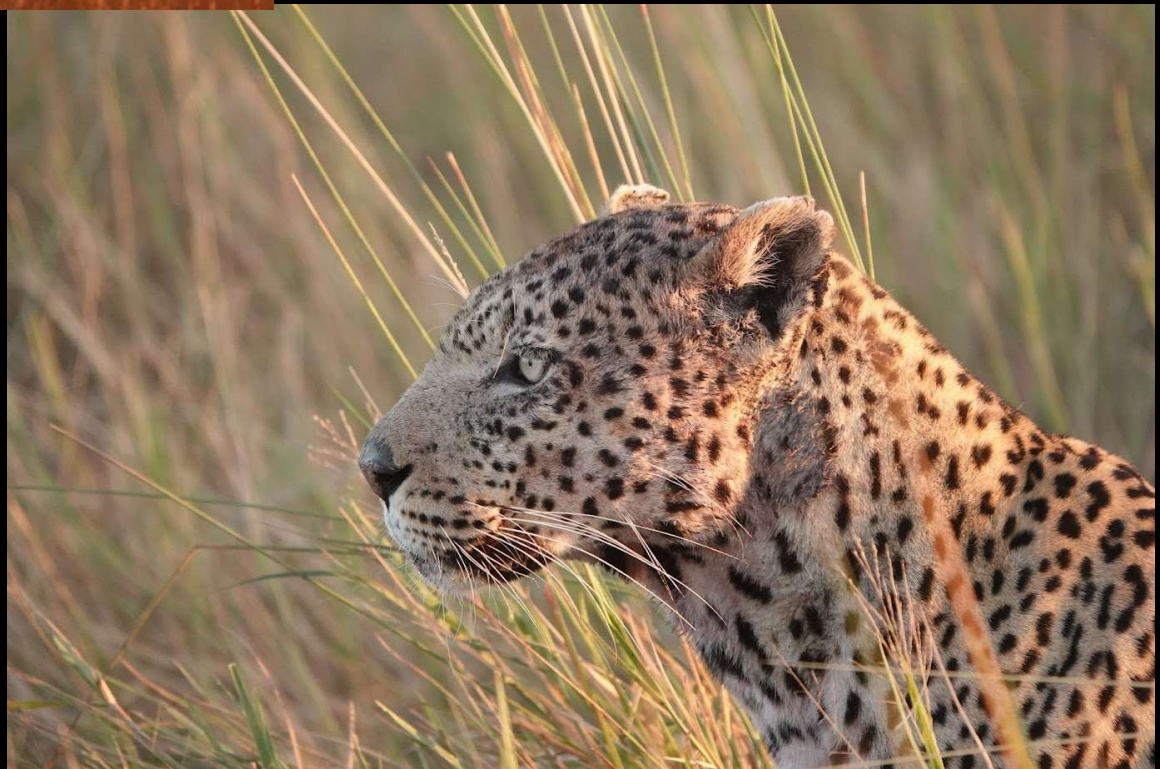
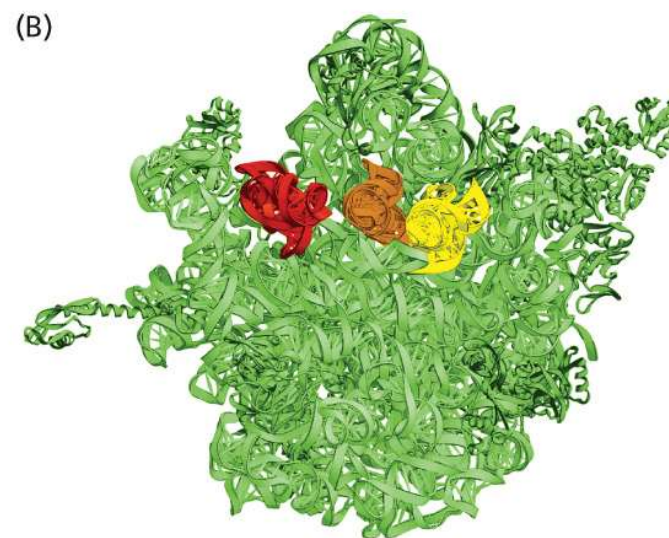
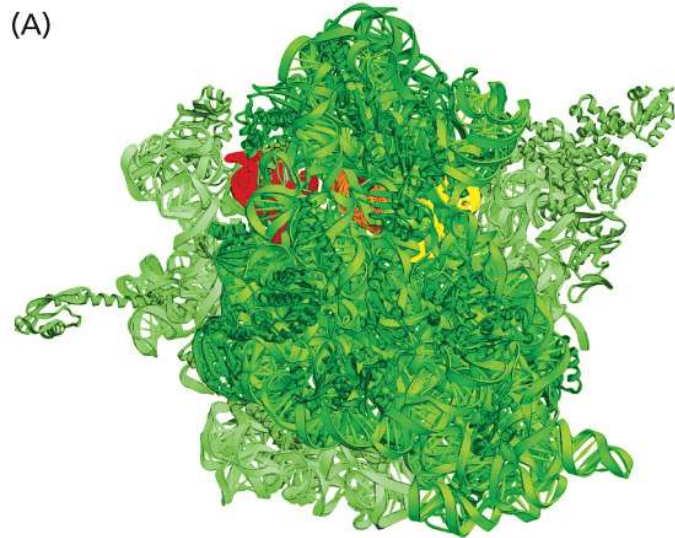


Figure 2-44

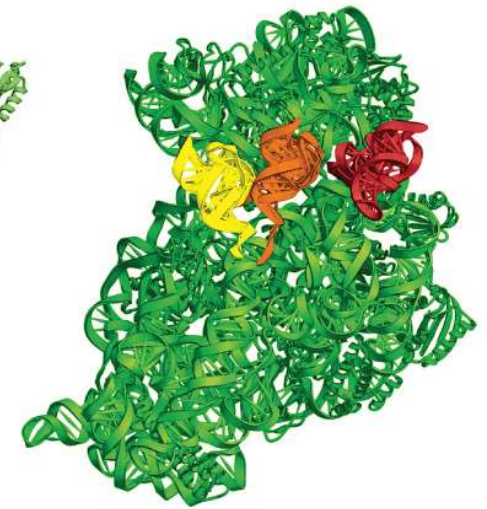


break

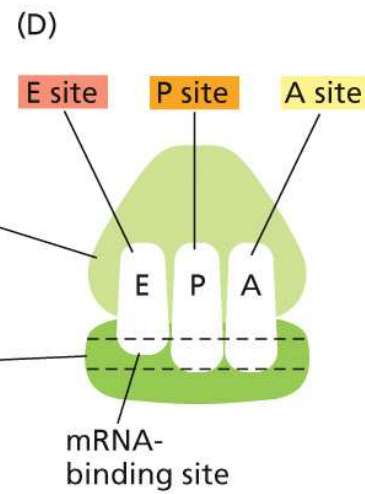
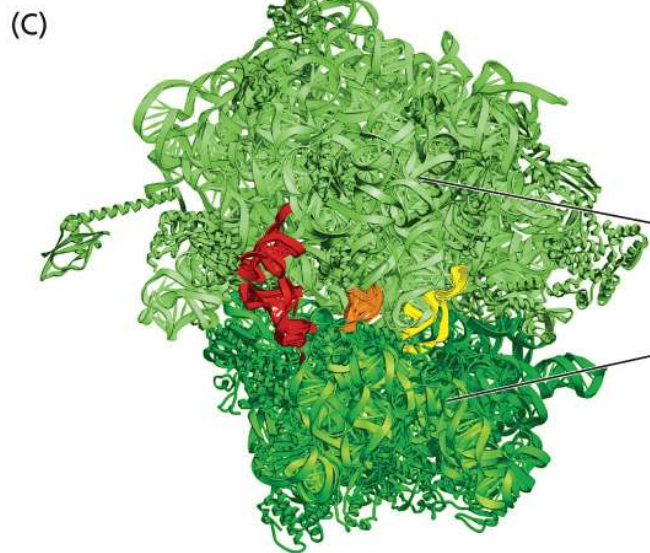


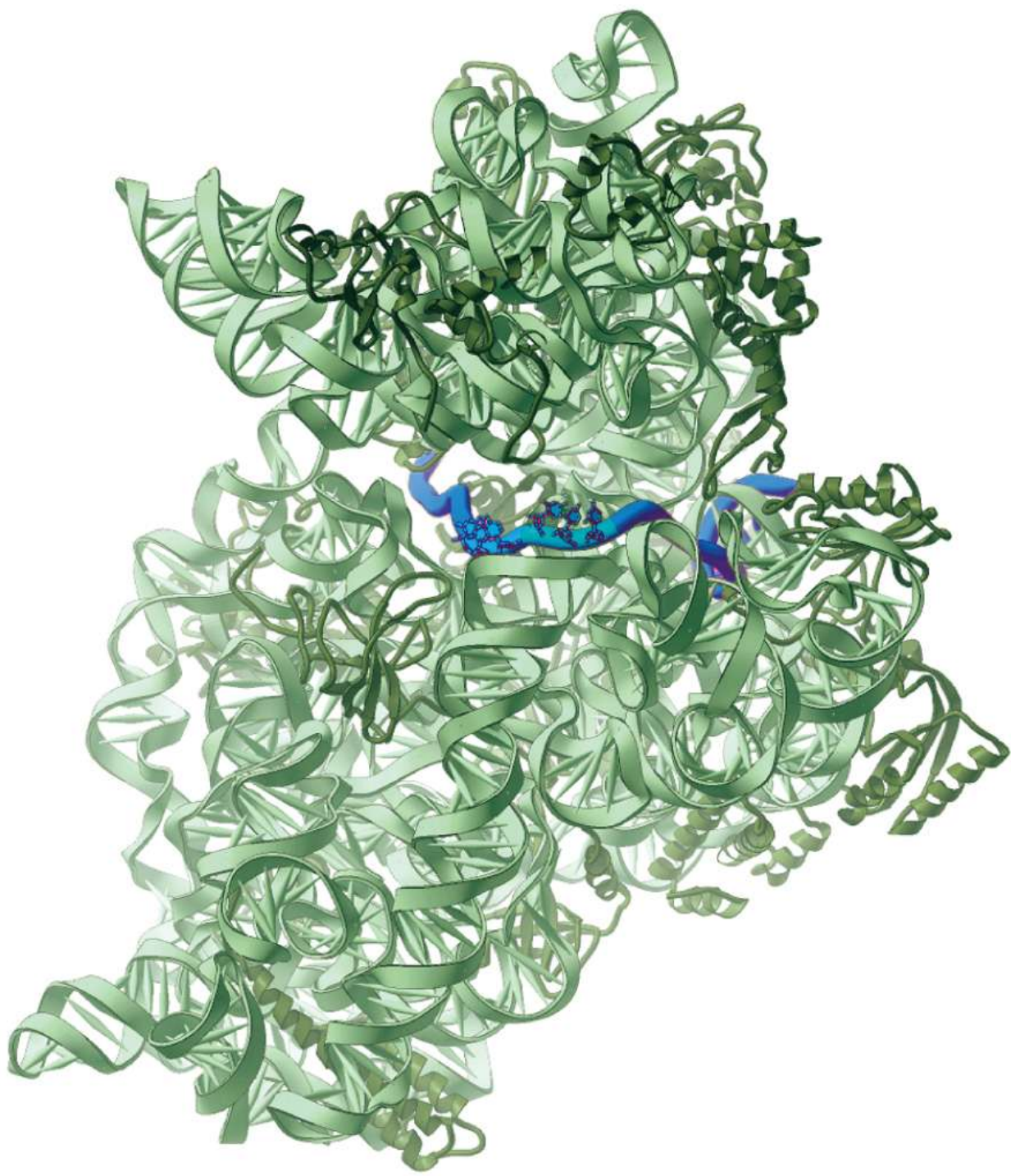


large subunit

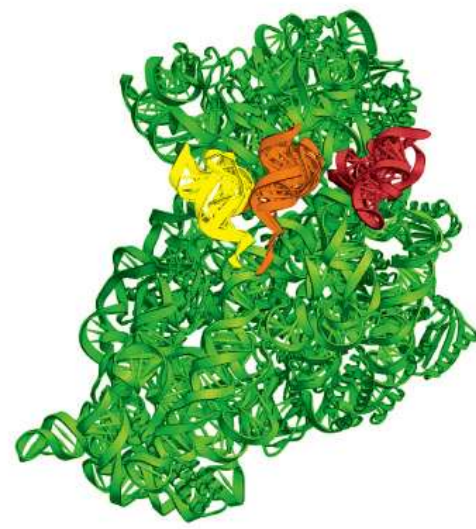


small subunit





Courtesy of Harry F. Noller, based on data in G.Z. Yusupova et al.,
Cell 106:233–241, 2001.

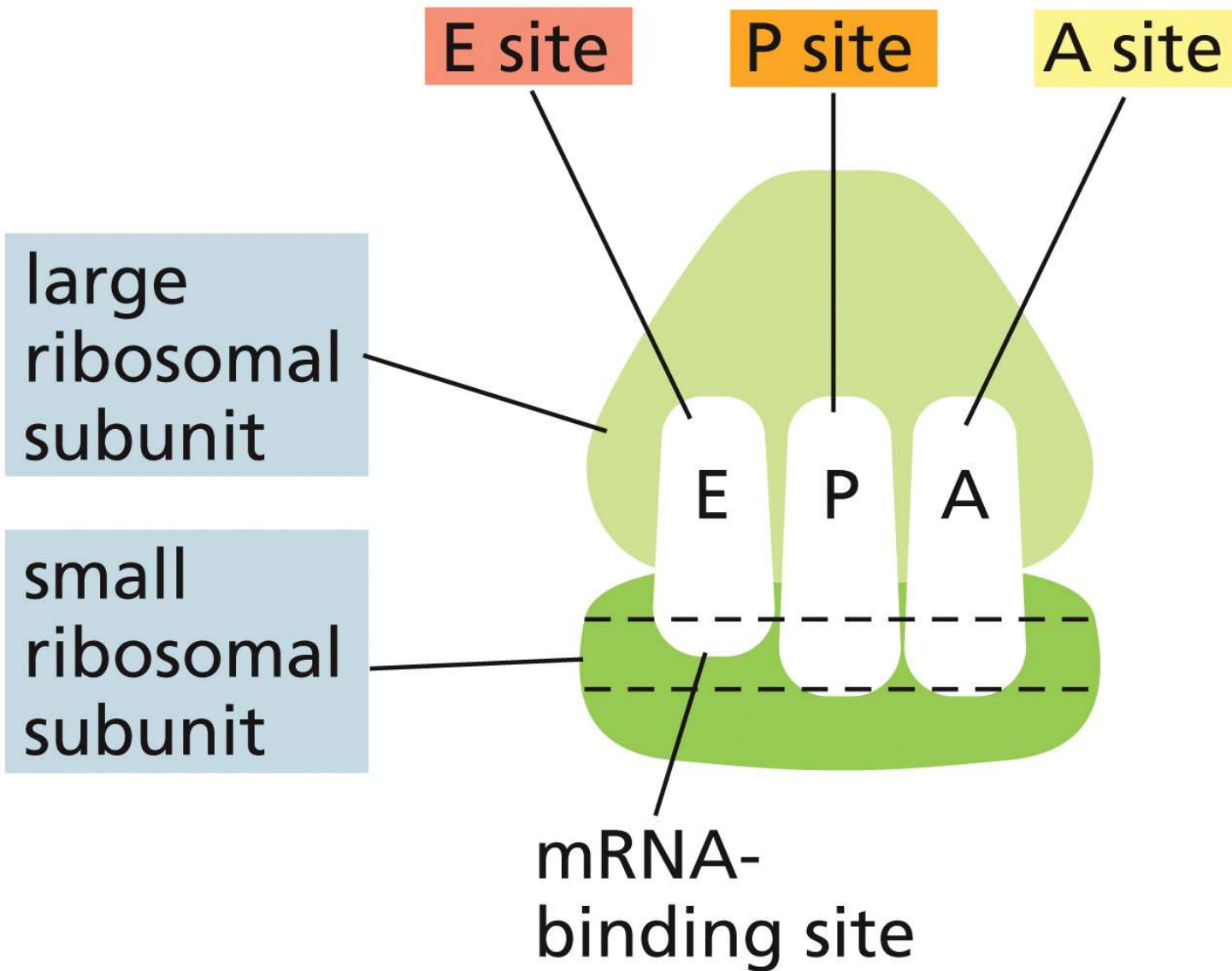


small subunit

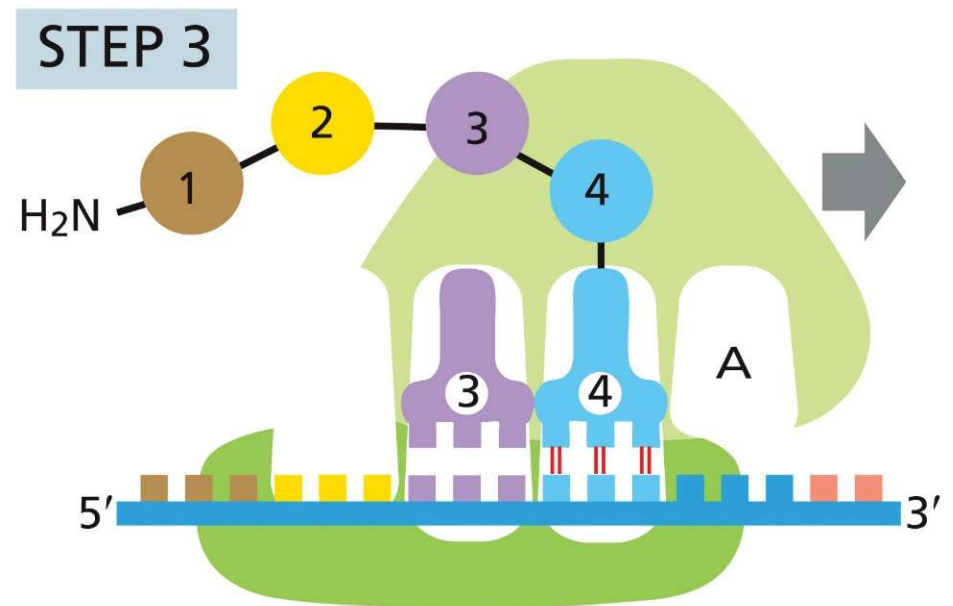
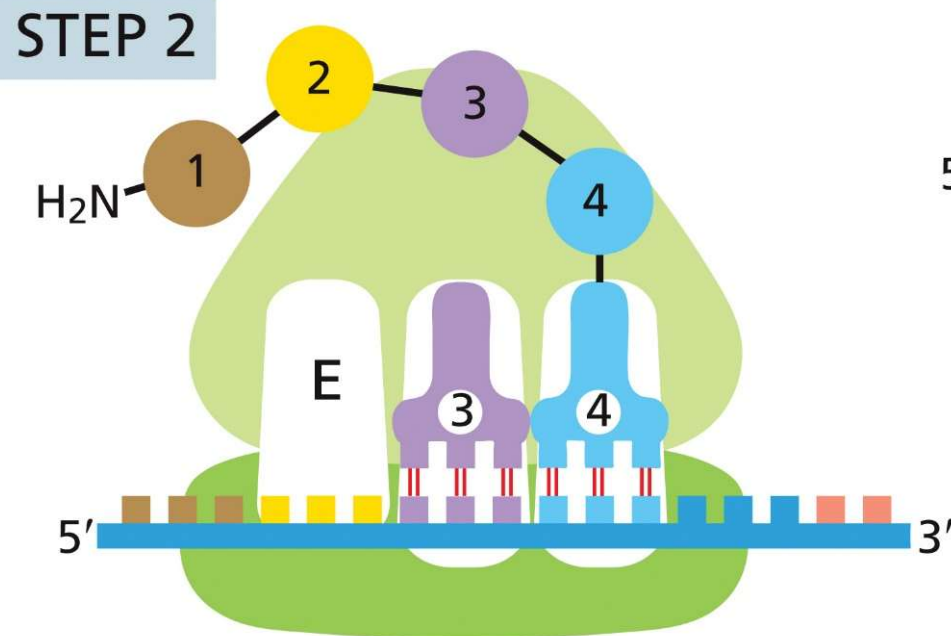
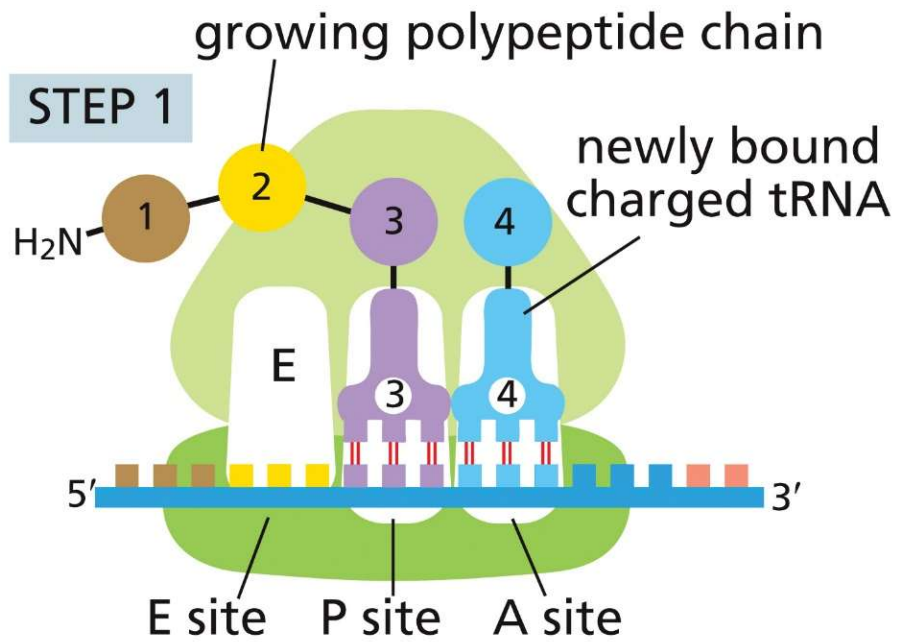
E: exit

P: peptidyl-tRNA

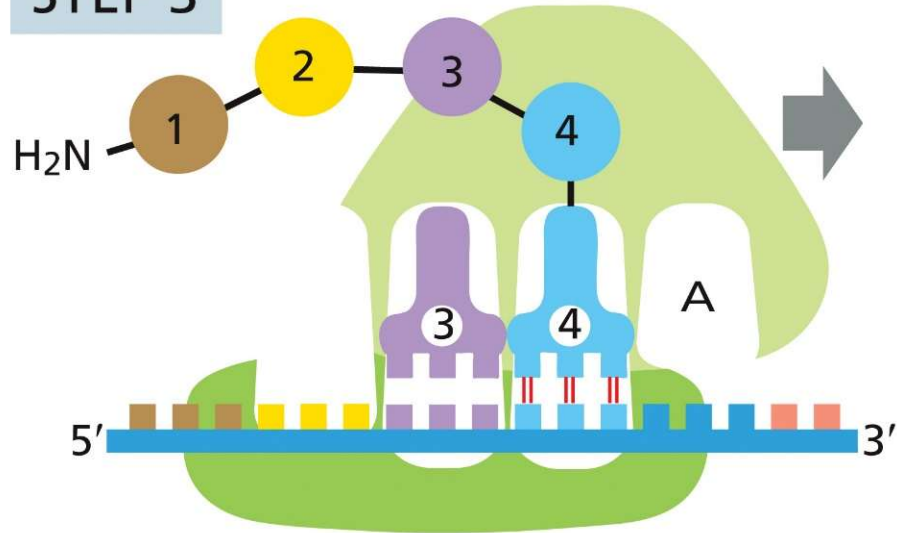
A: aminoacyl-tRNA



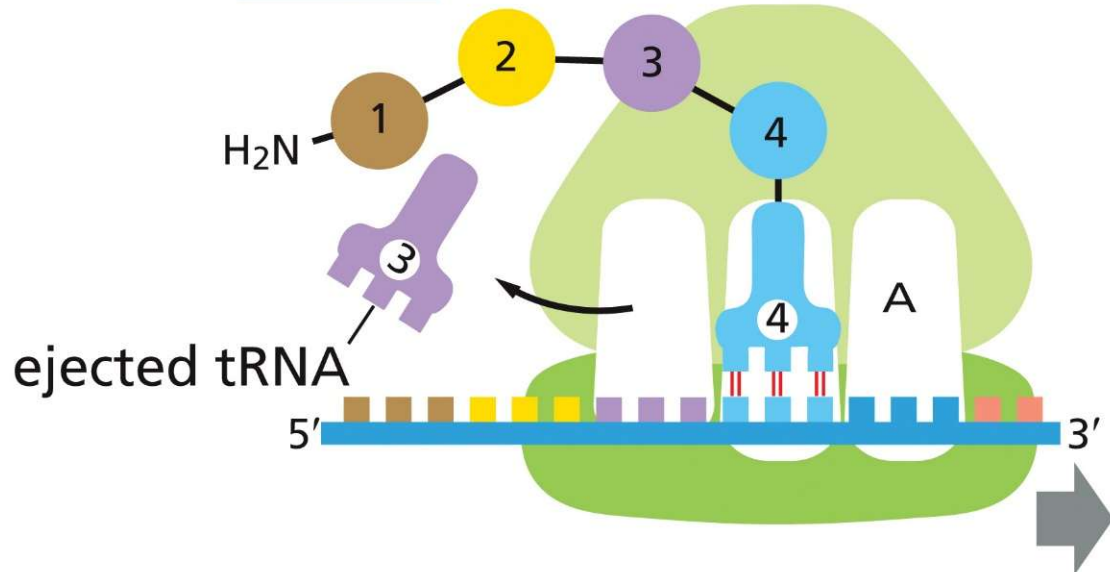
Elongation



STEP 3

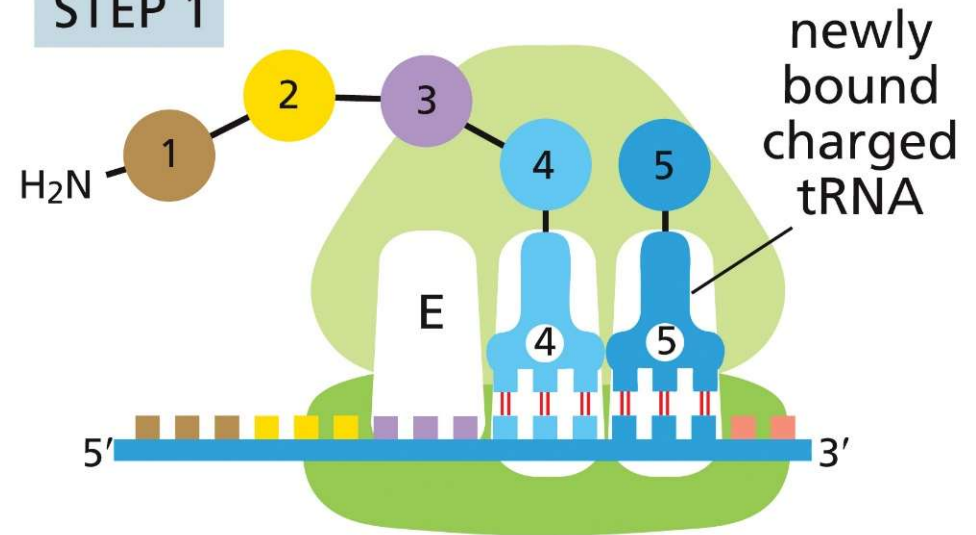


STEP 4



Elongation

STEP 1



Speed?

~ 6 aa/sec (mouse)

Average protein = 350-450 aa

Titin = 30.000 aa

Elongation factors (1): speed and accuracy

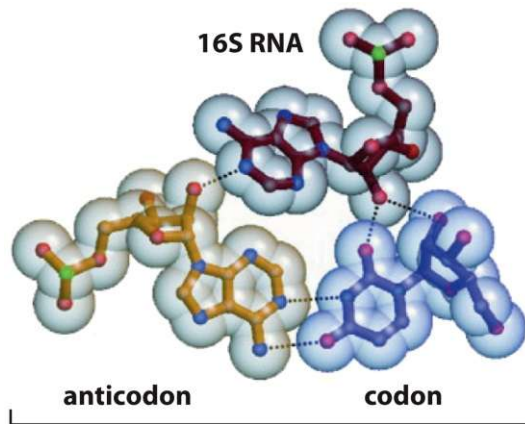
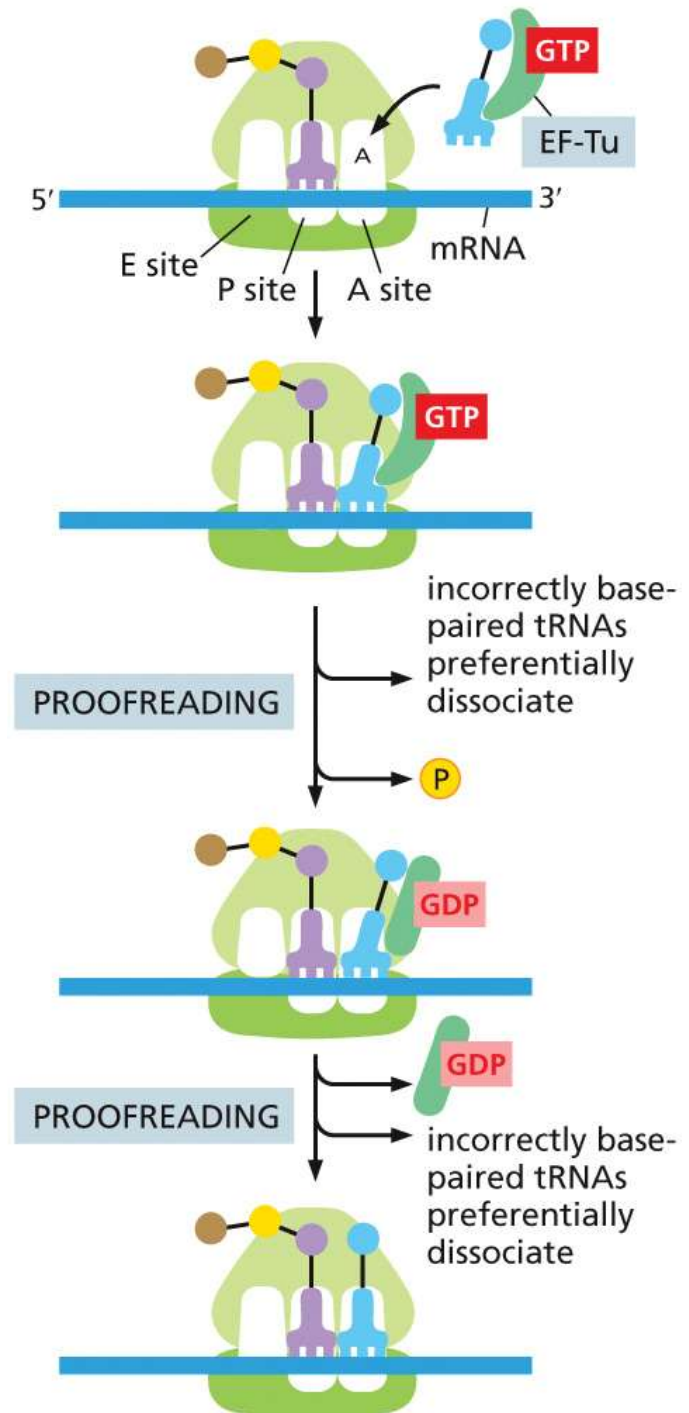
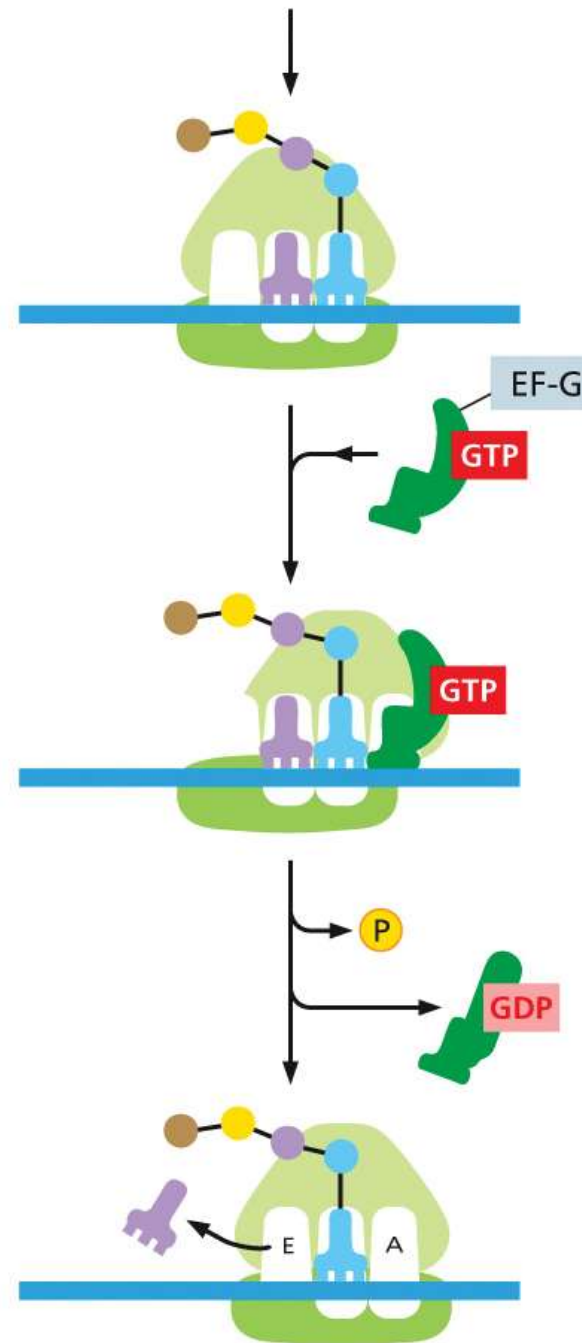
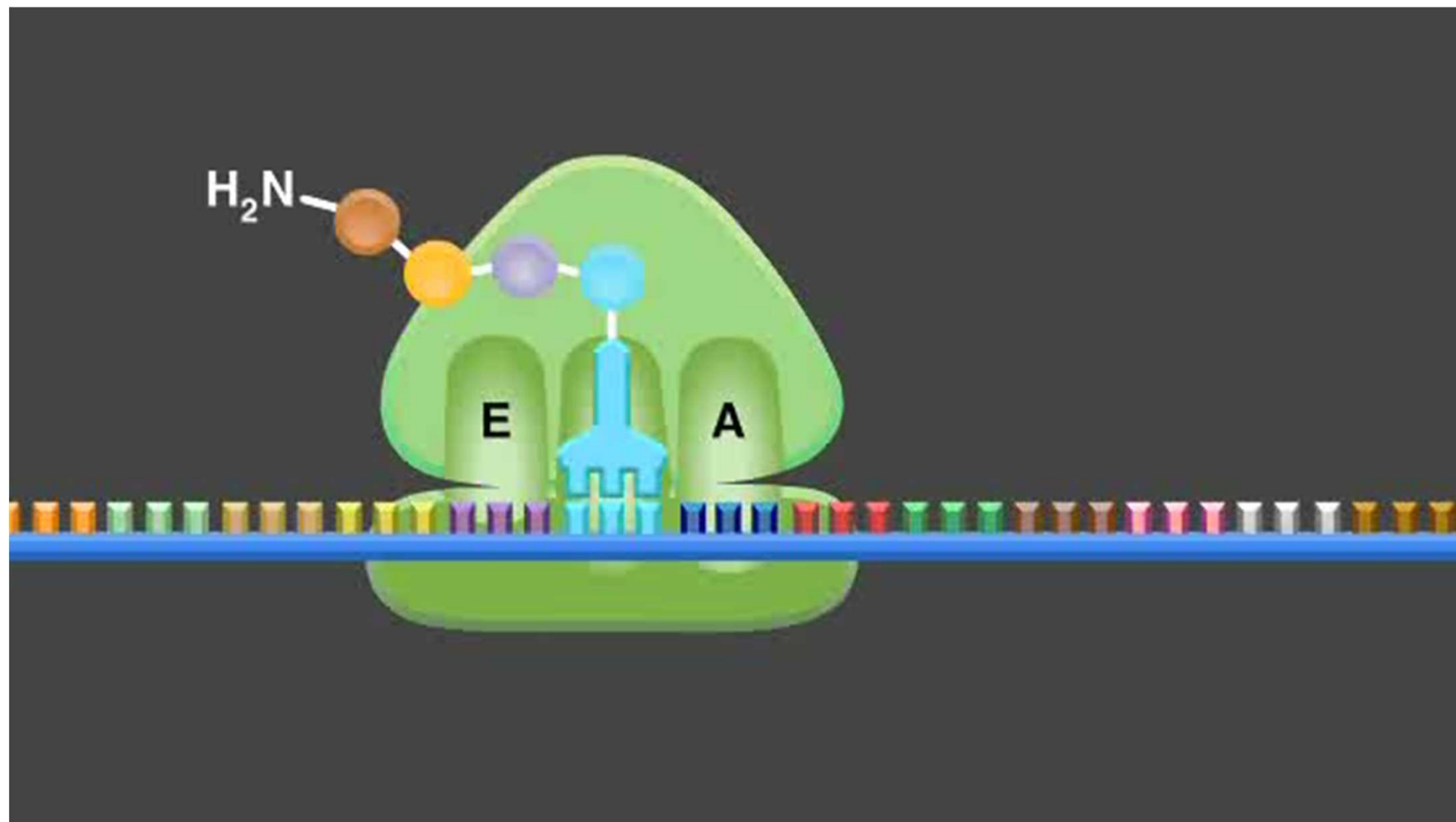


Figure 6-66 Molecular Biology of the Cell 6e (© Garland Science 2015)

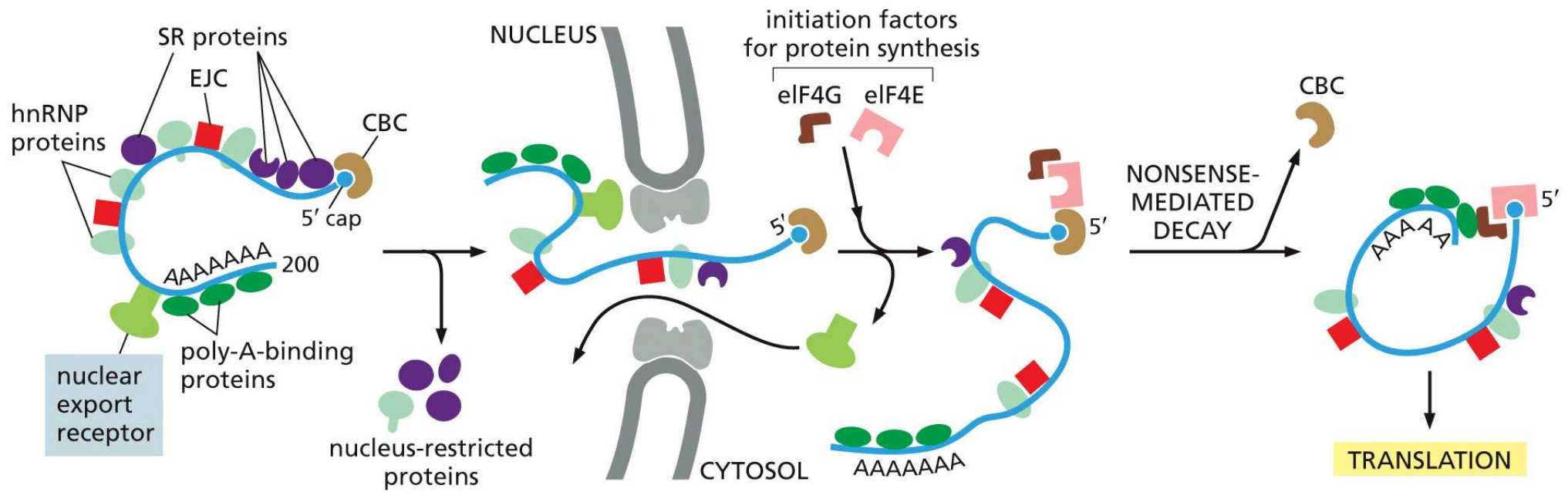


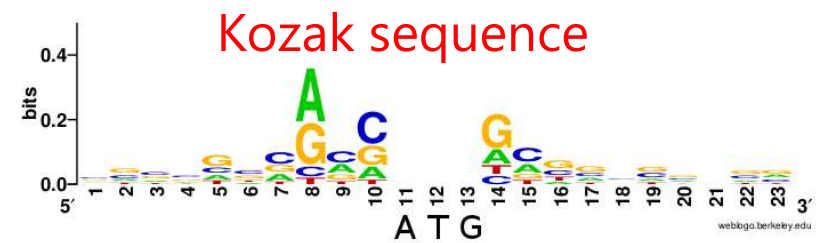
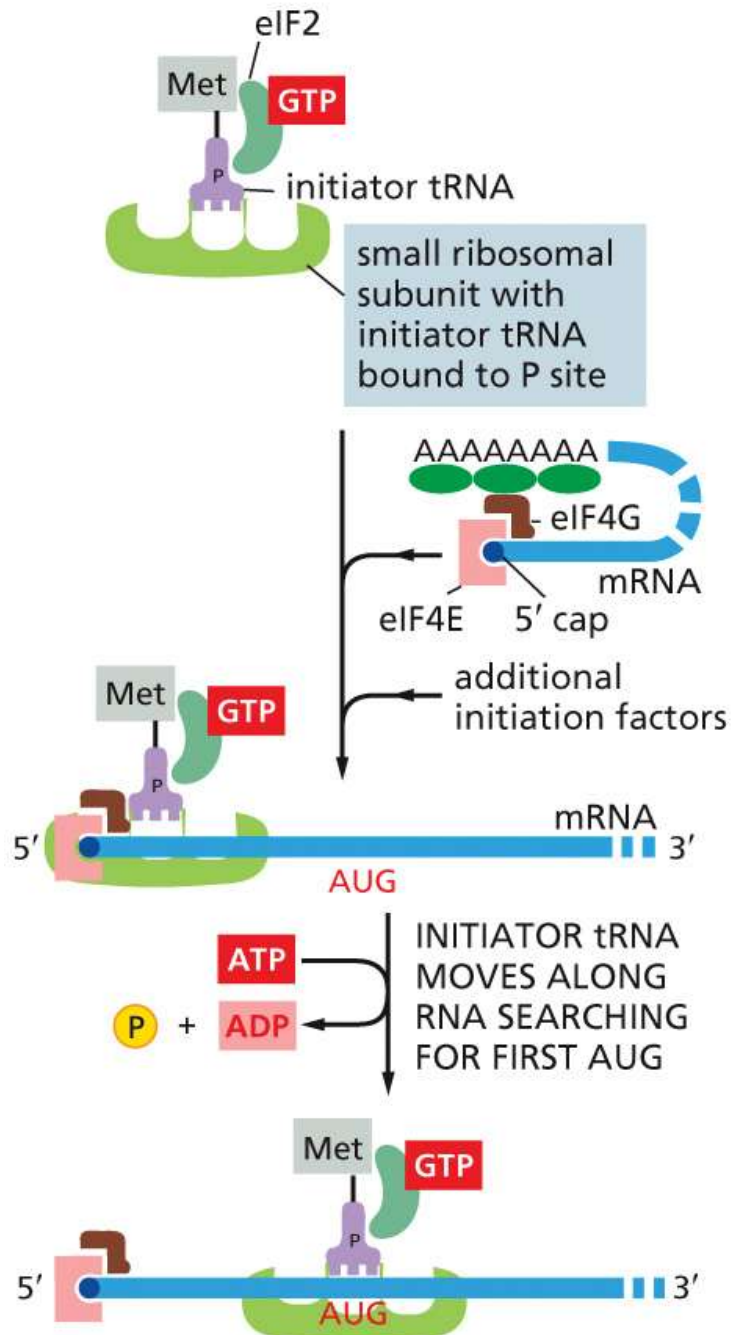
Elongation factors (2): efficiency

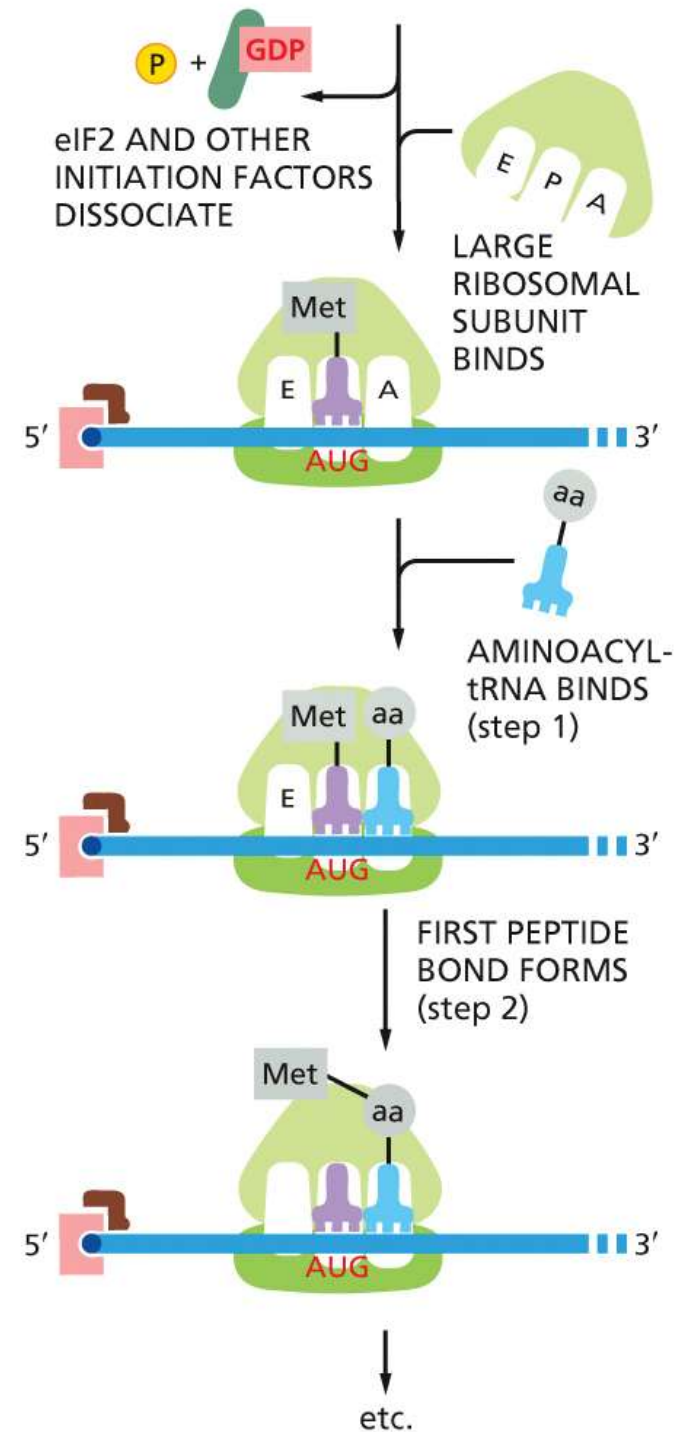
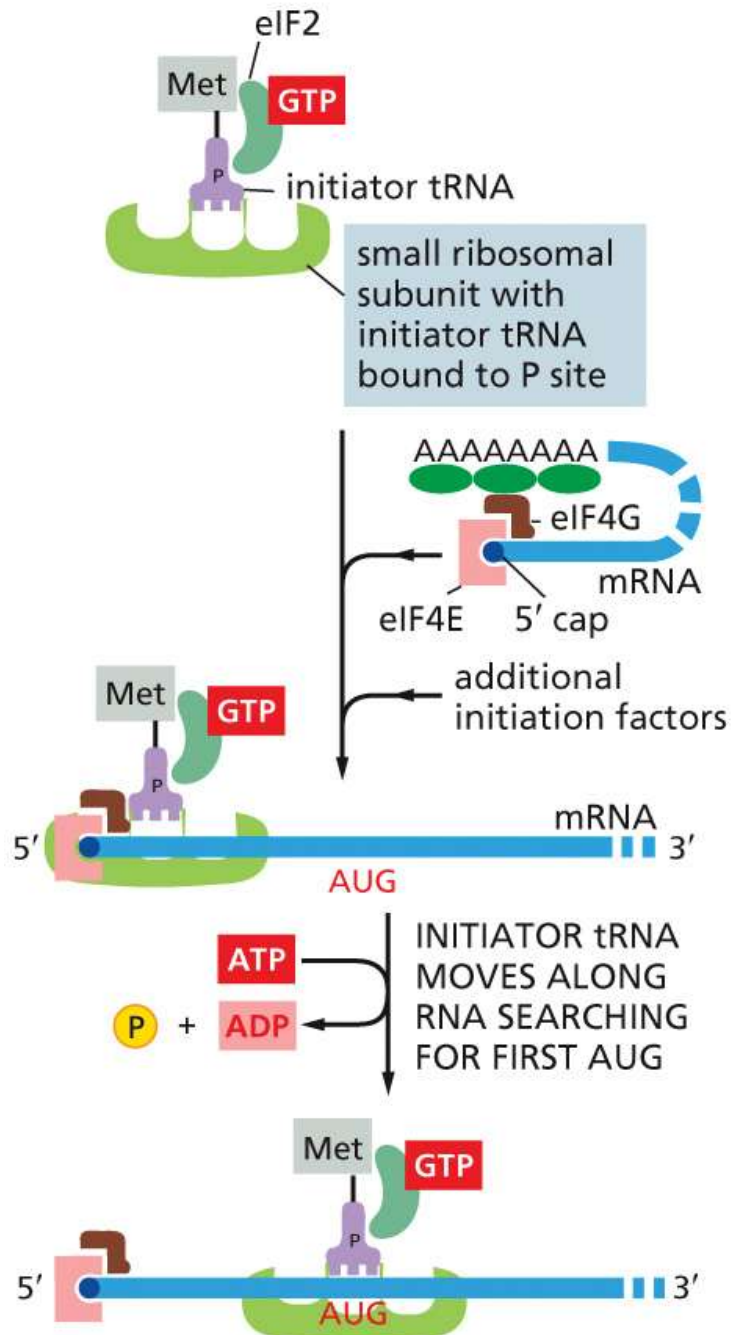




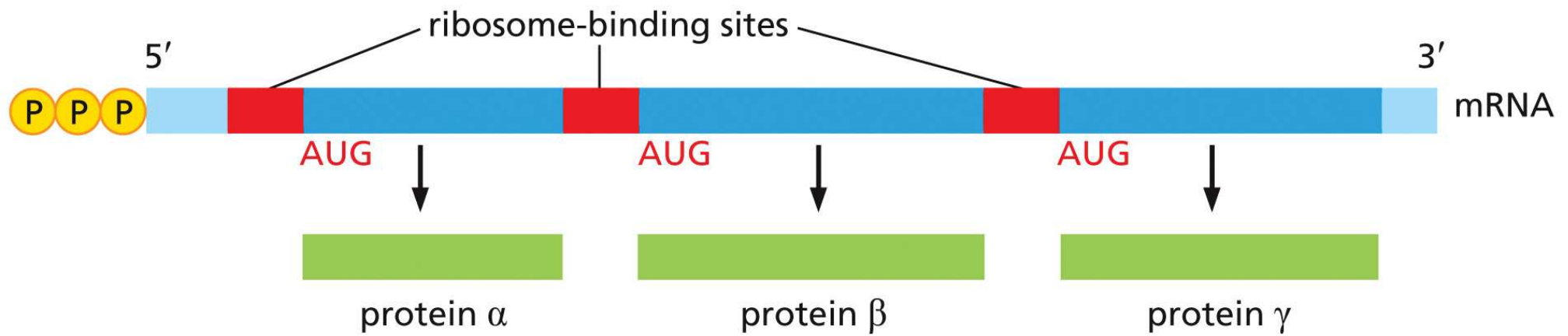
Initiation



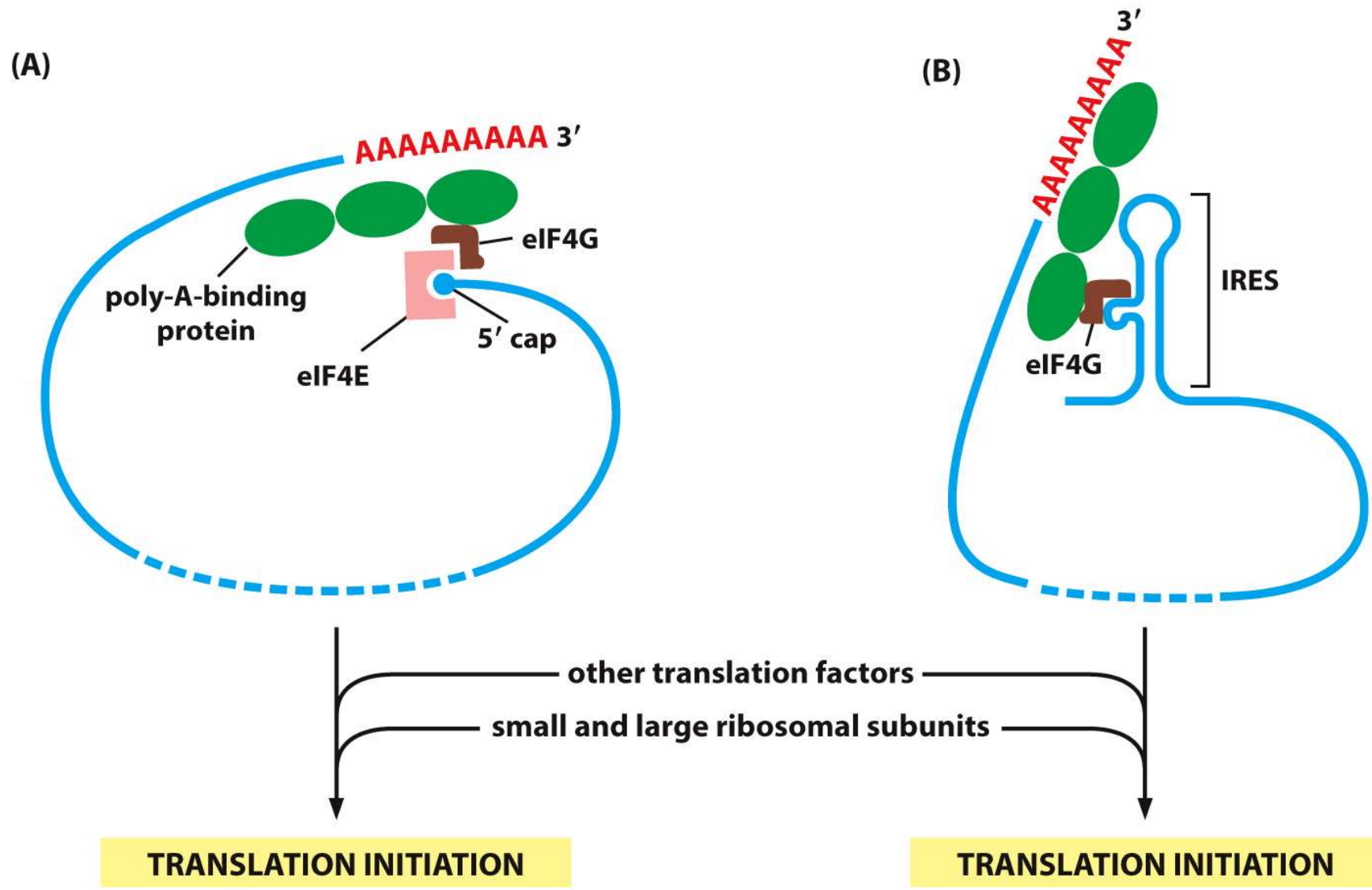




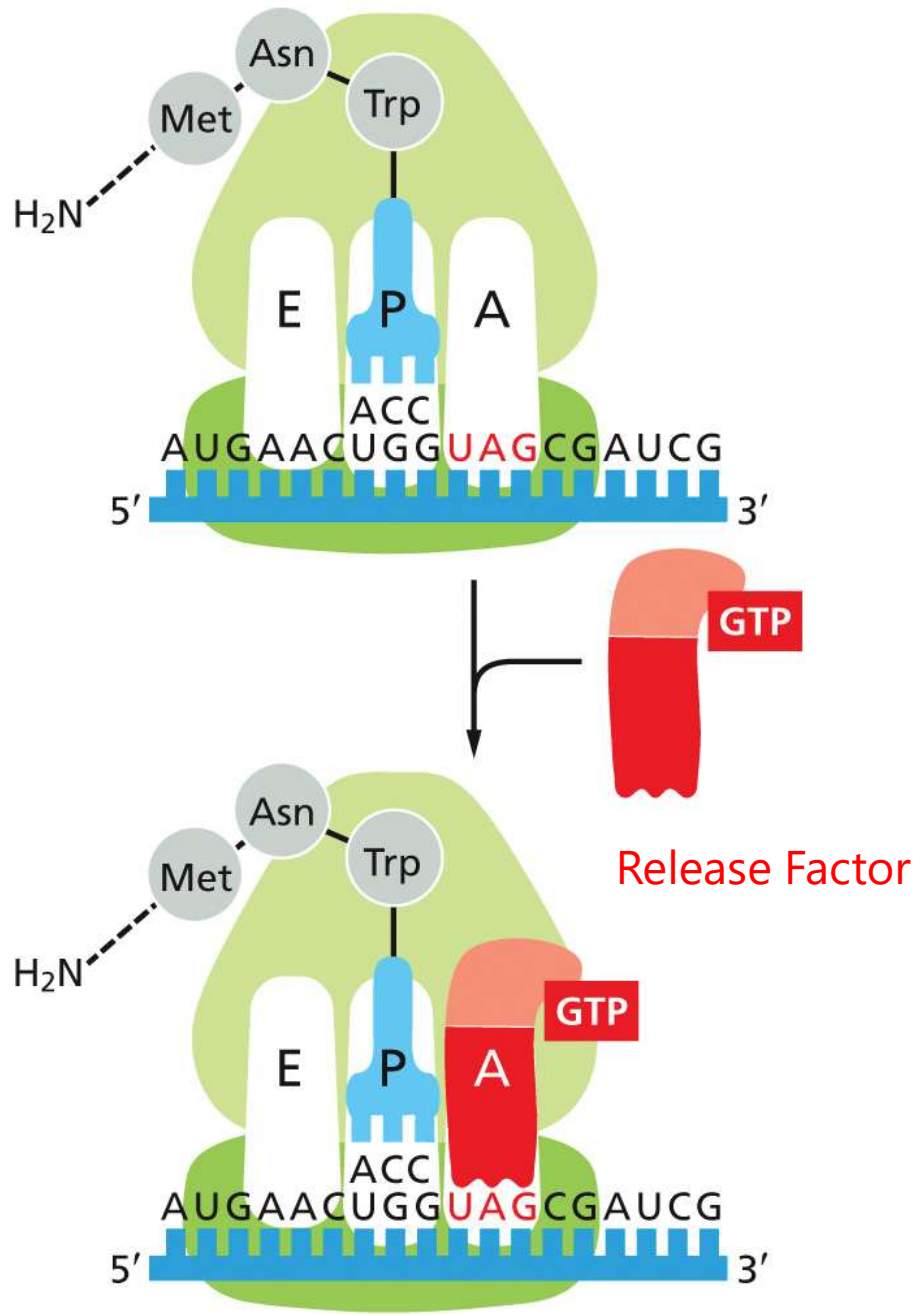
Initiation in bacteria: Shine-Dalgarno sequences



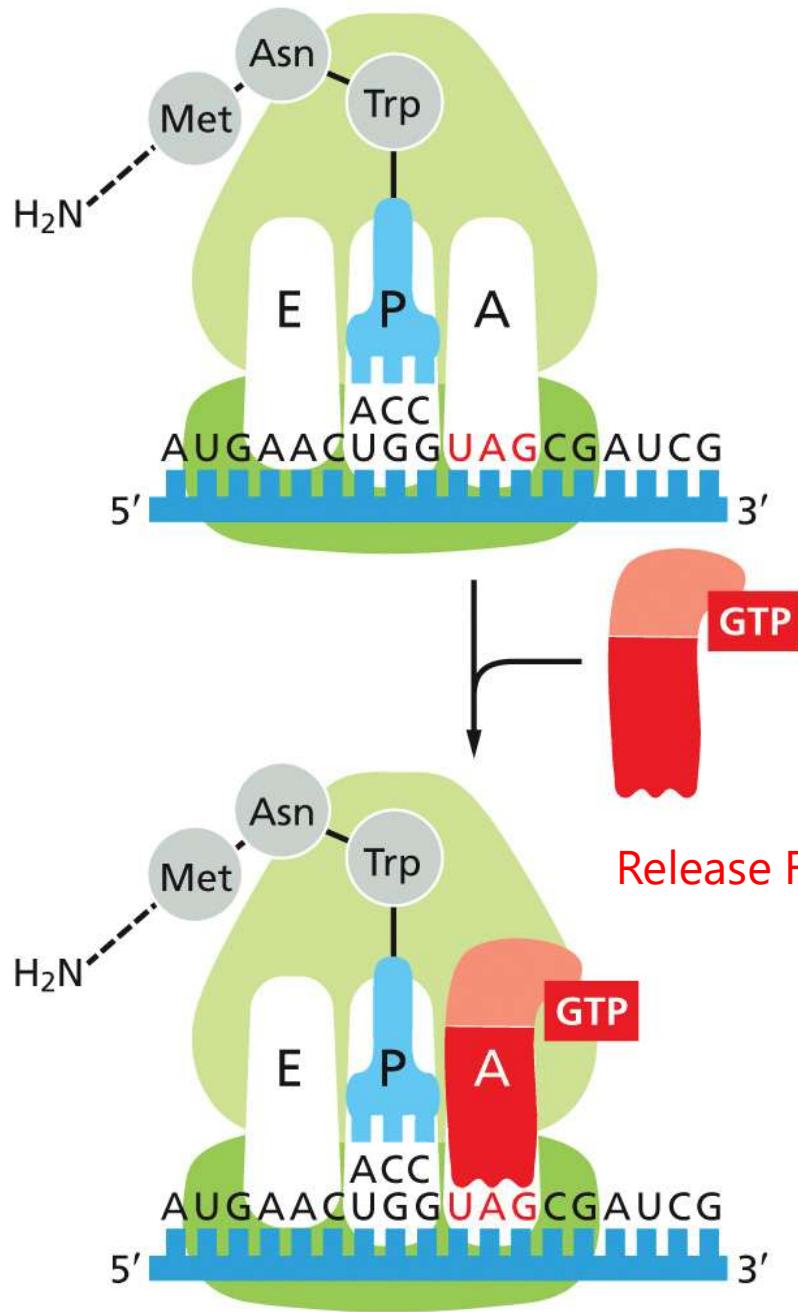
Alternative initiation: the IRES sequence (Ch. 7)



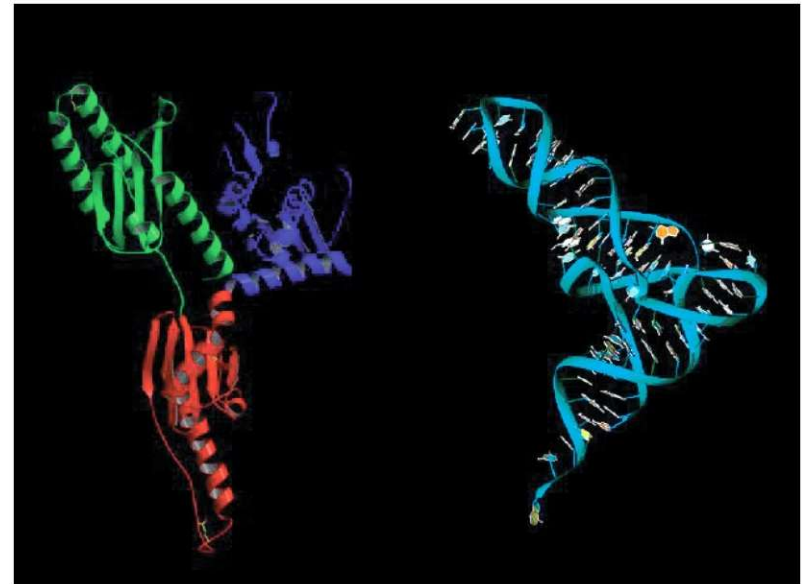
Termination

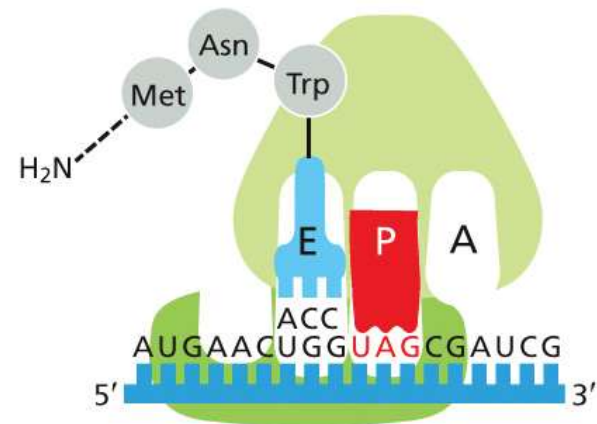
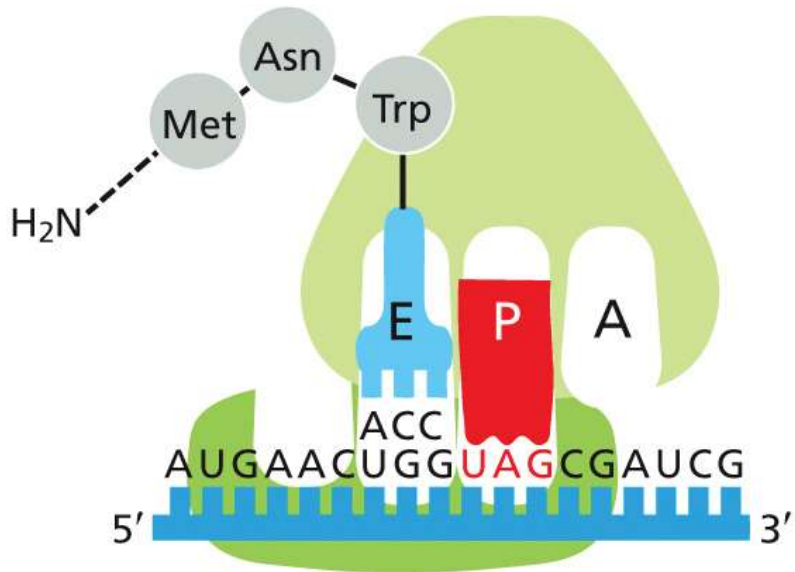
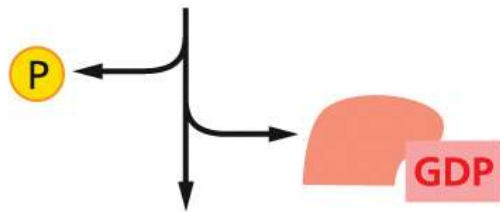
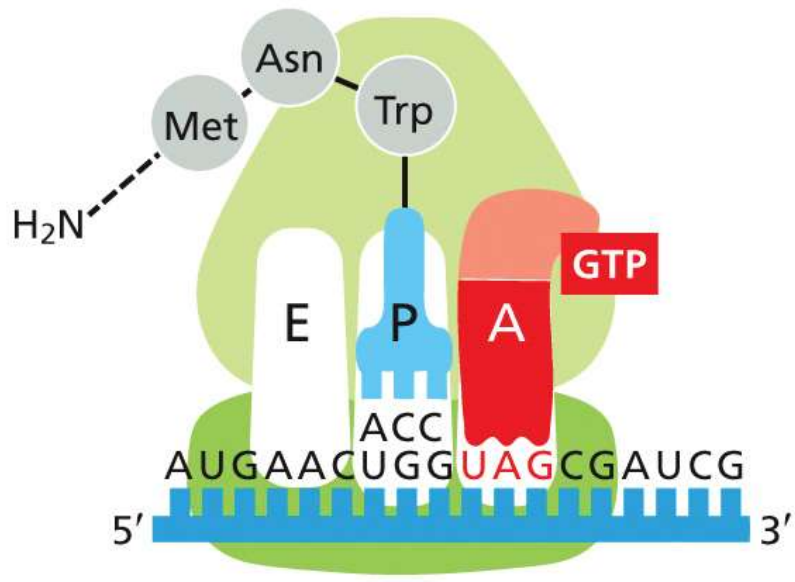


Termination

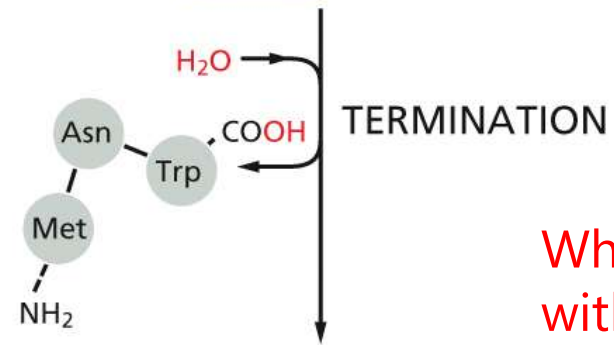


mimicry

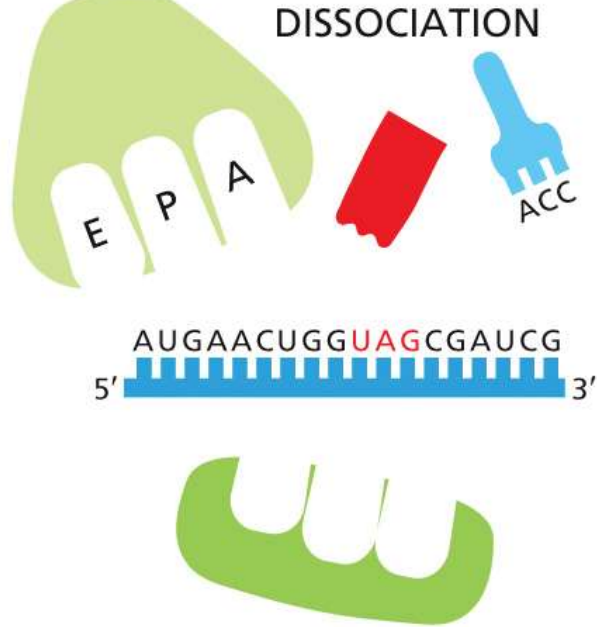


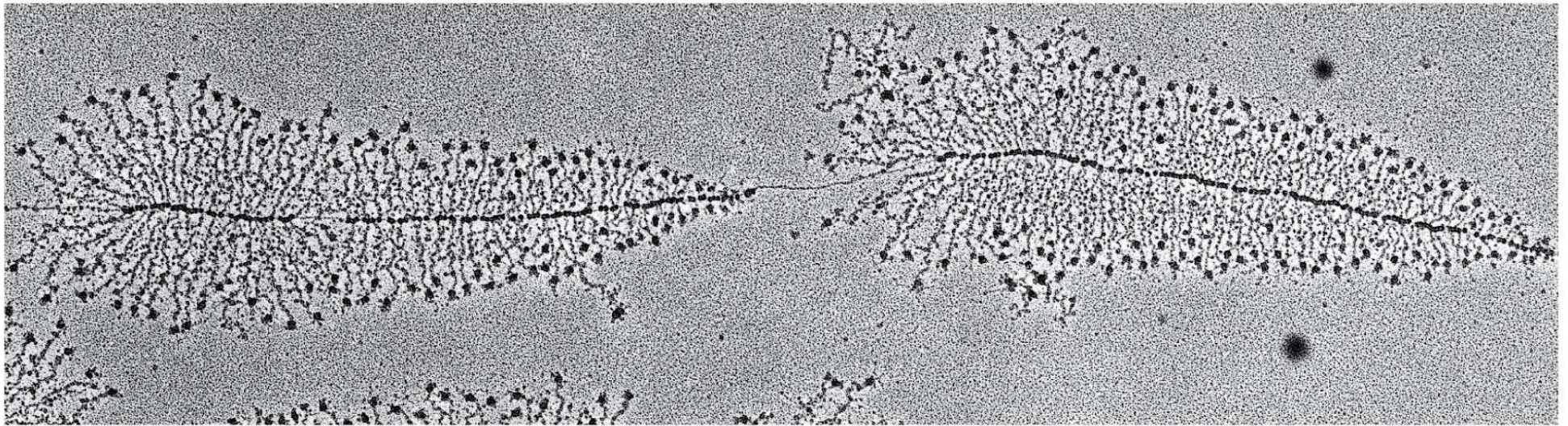


**addition
of H₂O!**



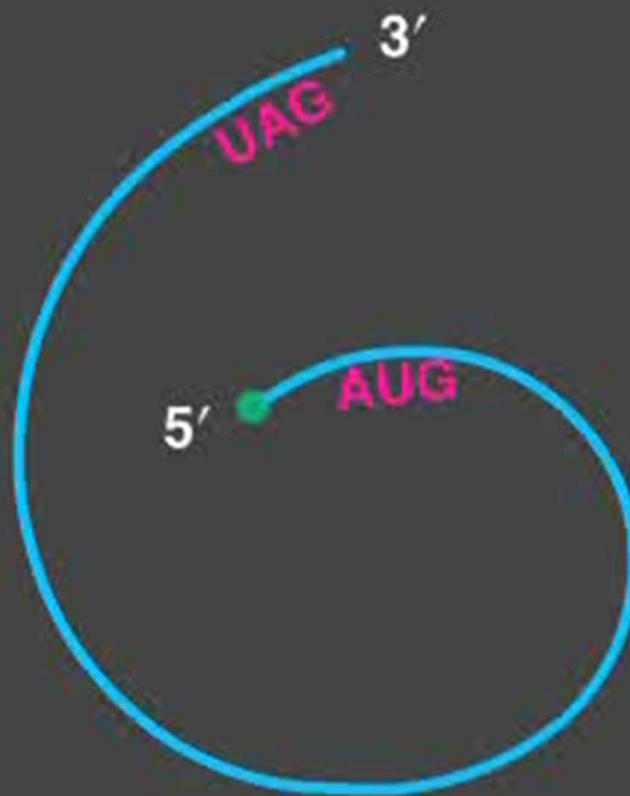
**What happens
with subunits?**



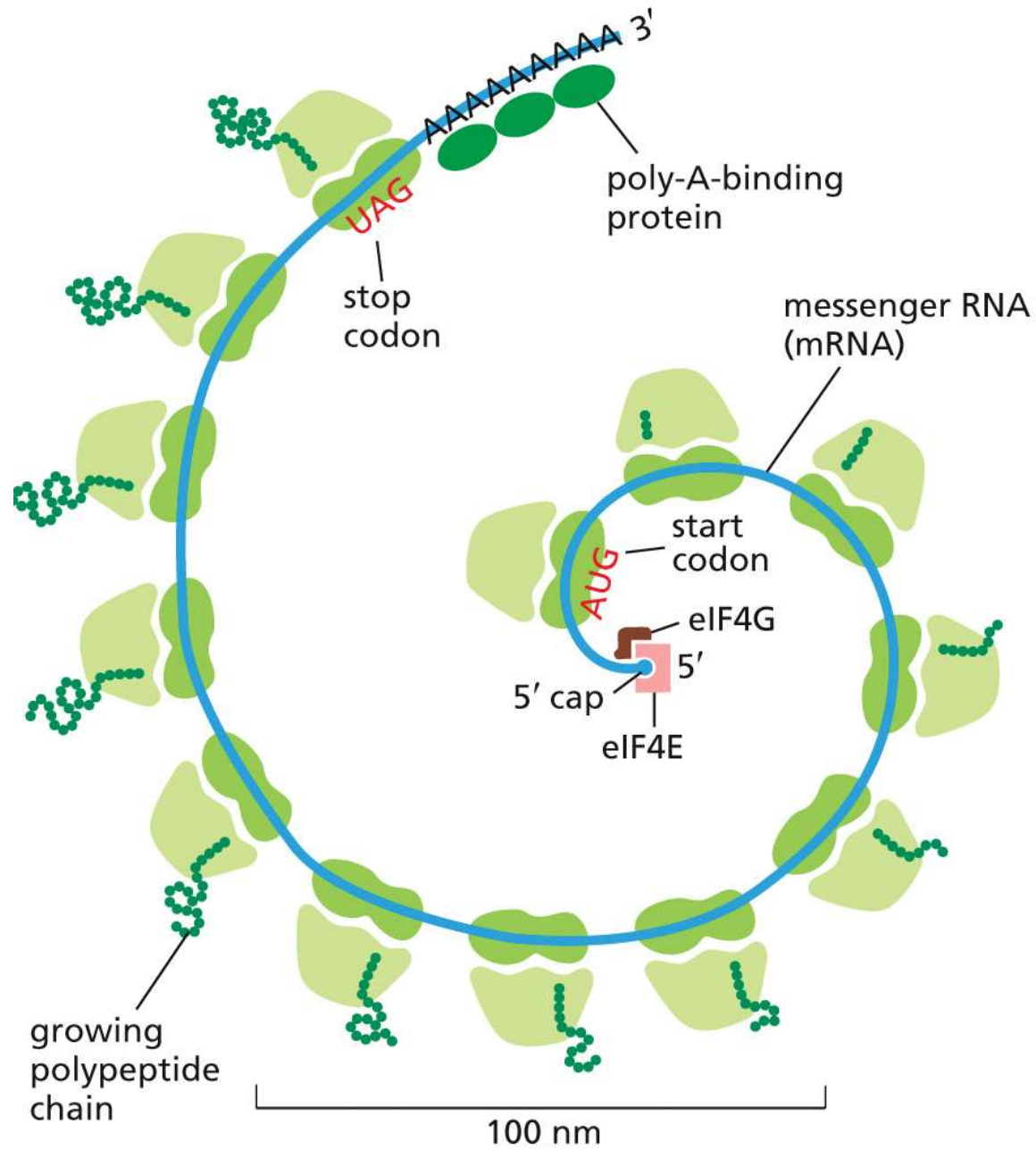


Courtesy of Ulrich Scheer

1 μm

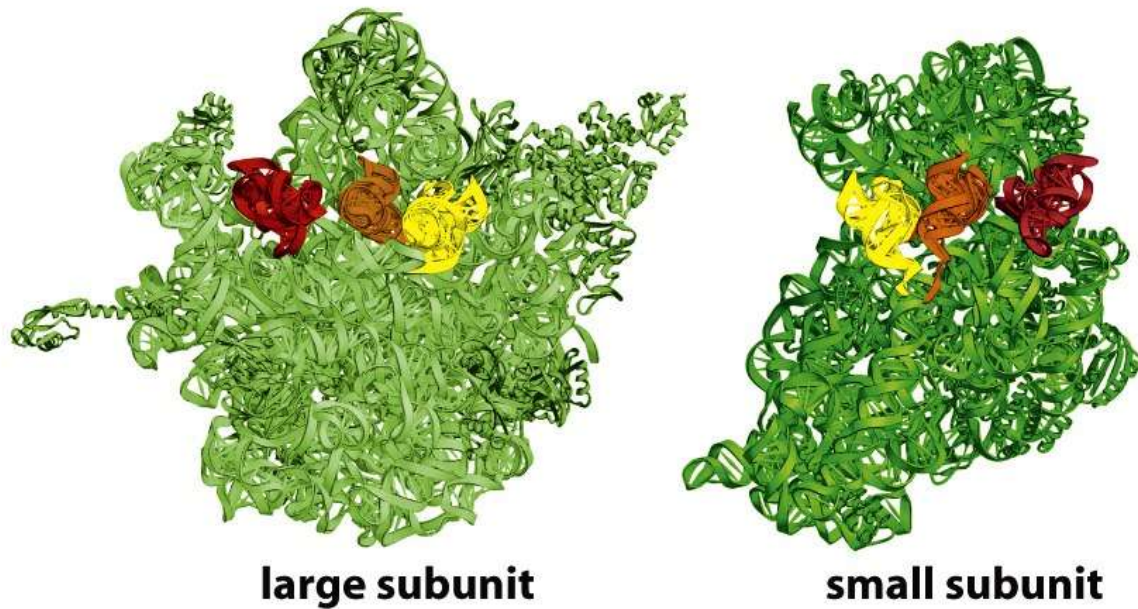


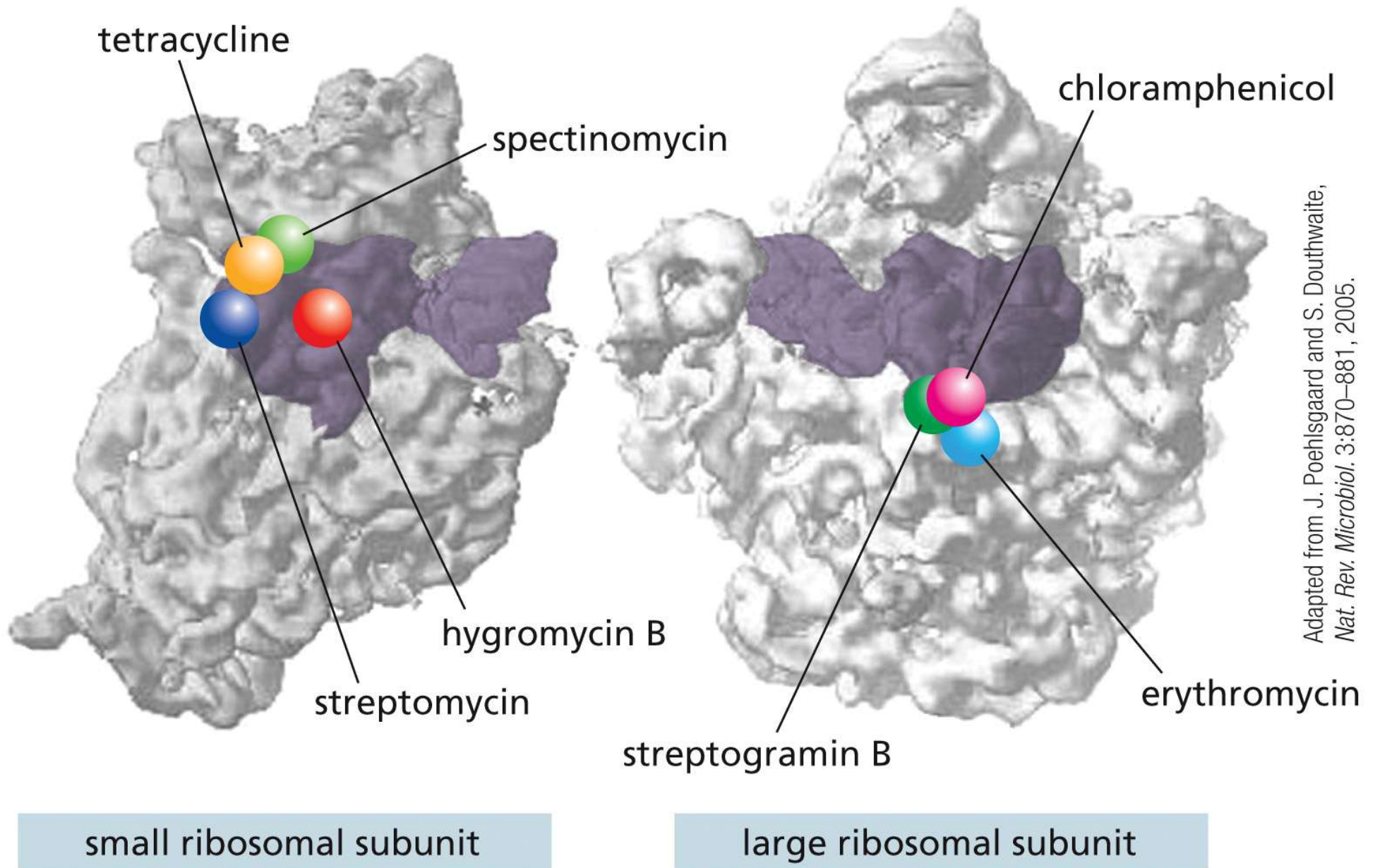
Polyribosome



What is the connection between translation and antibiotics?

Fungi vs. Bacteria → coevolution → antibiotics



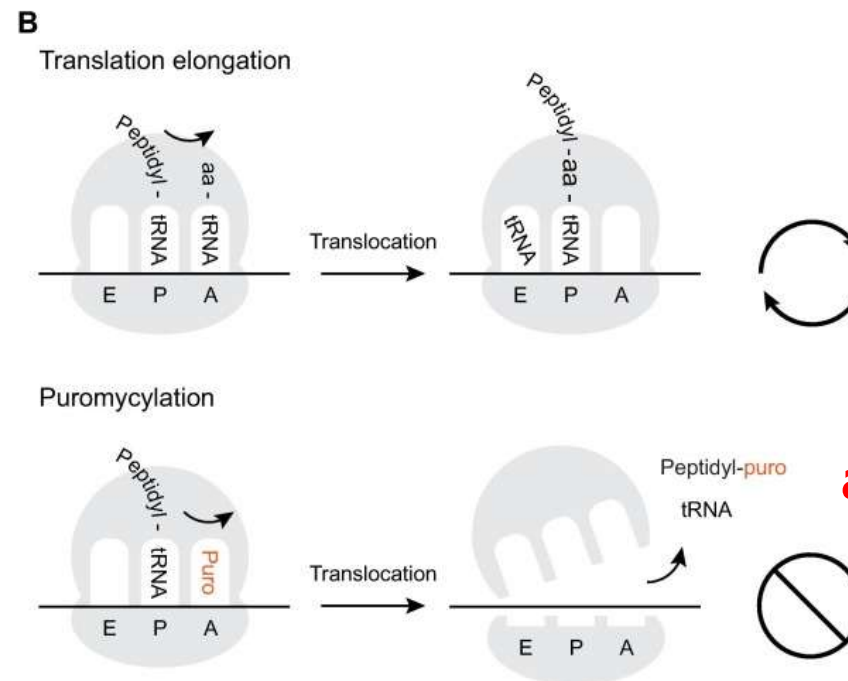
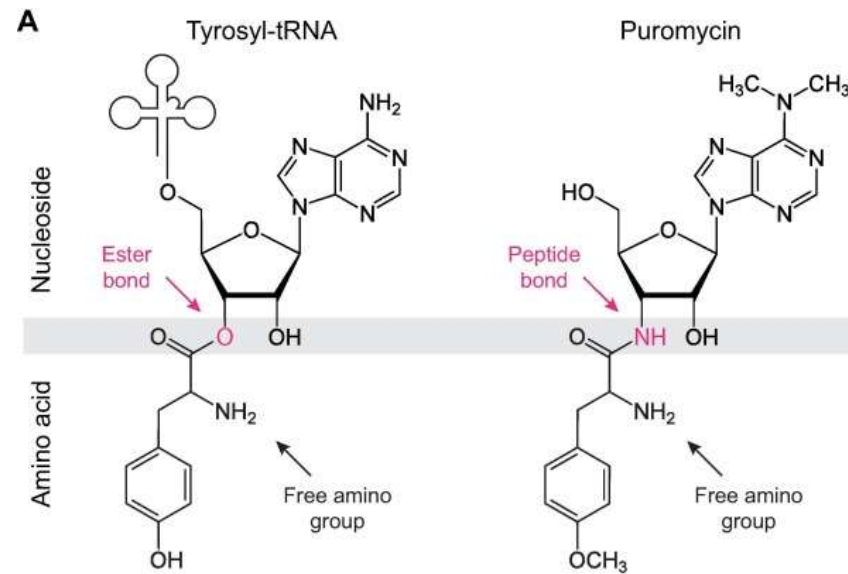


Adapted from J. Poehlisgaard and S. Douthwaite,
Nat. Rev. Microbiol. 3:870–881, 2005.

TABLE 6–4 Inhibitors of Protein or RNA Synthesis

Inhibitor	Specific effect
Acting only on bacteria	
Tetracycline	Blocks binding of aminoacyl-tRNA to the A site of the ribosome
Streptomycin	Prevents the transition from translation initiation to chain elongation and also causes miscoding
Chloramphenicol	Blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–68)
Erythromycin	Binds in the exit tunnel of the ribosome and thereby inhibits elongation of the peptide chain
Rifamycin	Blocks initiation of RNA chains by binding to RNA polymerase (prevents RNA synthesis)
Acting on bacteria and eukaryotes	
Puromycin	Causes the premature release of nascent polypeptide chains by its addition to the growing chain end
Actinomycin D	Binds to DNA and blocks the movement of RNA polymerase (prevents RNA synthesis)
Acting on eukaryotes but not bacteria	
Harringtonine	Blocks the A site of the 60S ribosome subunit after translation initiation, but before elongation (see Figure 6–74)
Anisomycin	Blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–68)
α -Amanitin	Blocks mRNA synthesis by binding preferentially to RNA polymerase II
The ribosomes of eukaryotic mitochondria (and chloroplasts) often resemble those of bacteria in their sensitivity to inhibitors. Therefore, some of these antibiotics can have a deleterious effect on human mitochondria.	

Puromycin



addition of puromycin!

<https://doi.org/10.1016/j.csbj.2020.04.014>

Quality control: cap-polyA (1)

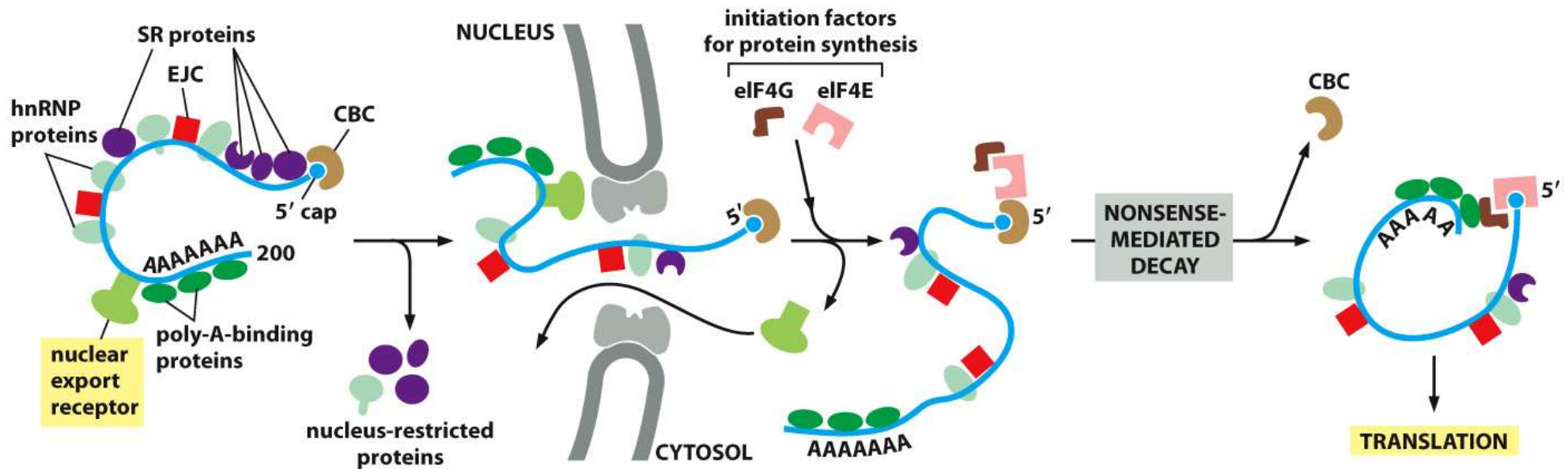
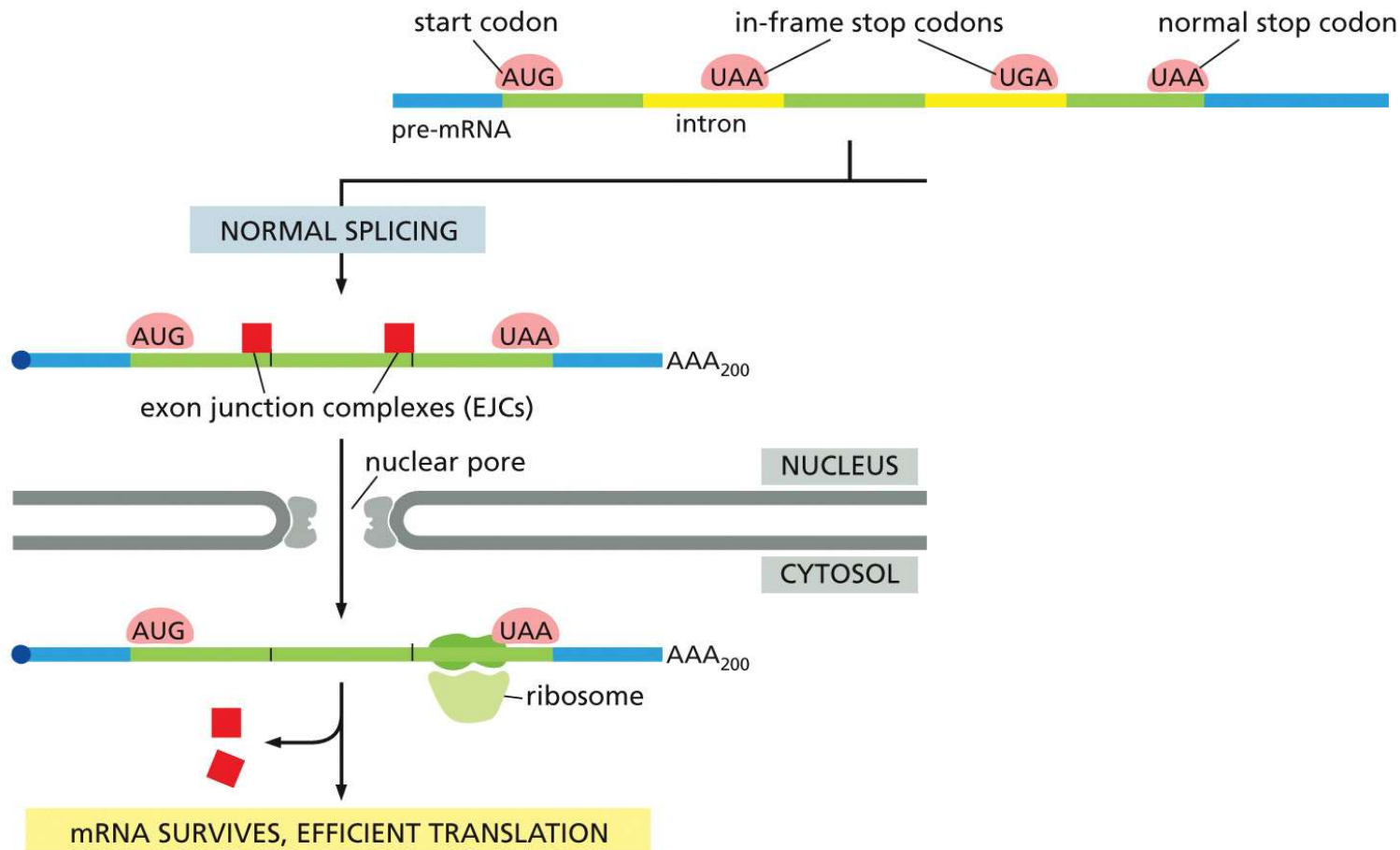


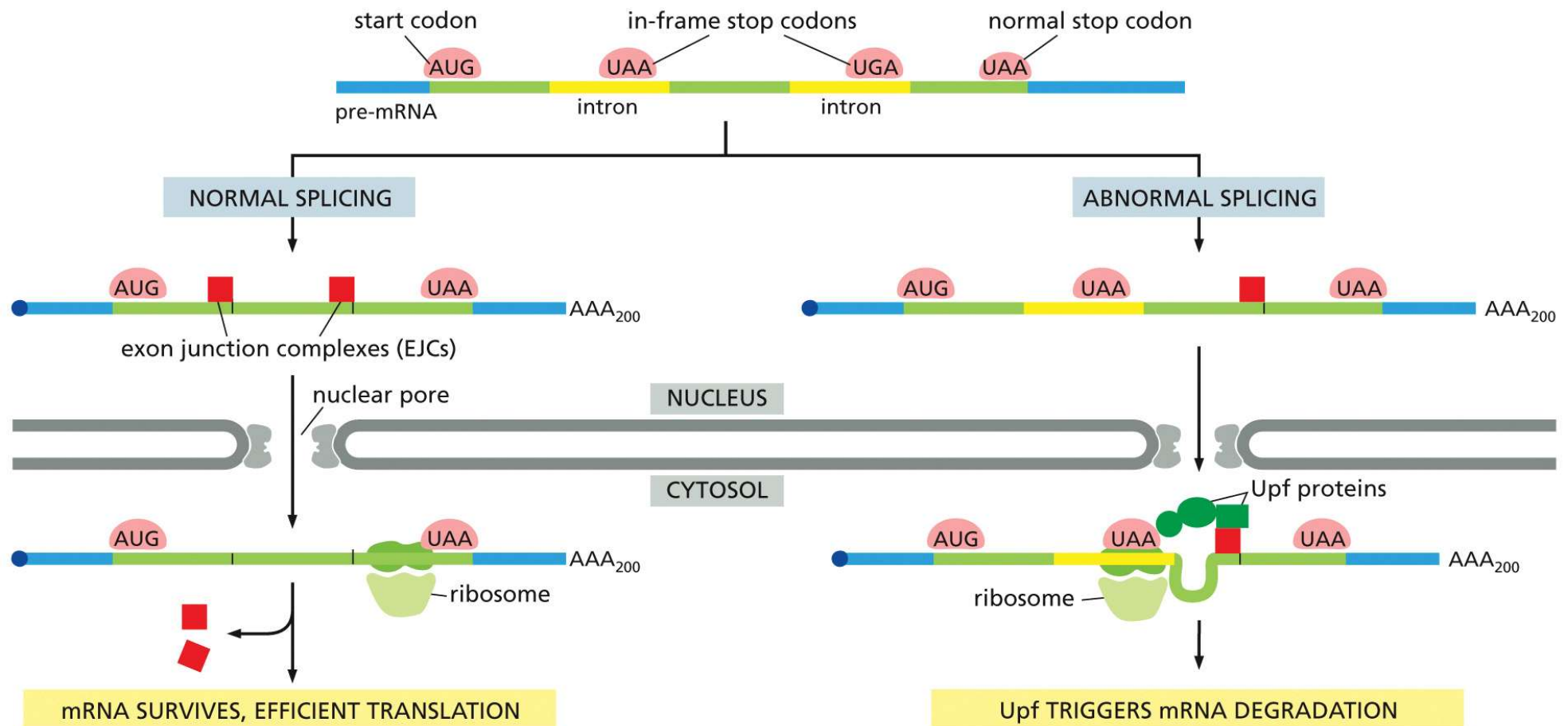
Figure 6-38 Molecular Biology of the Cell 6e (© Garland Science 2015)

Quality control: nonsense-mediated mRNA decay (2)



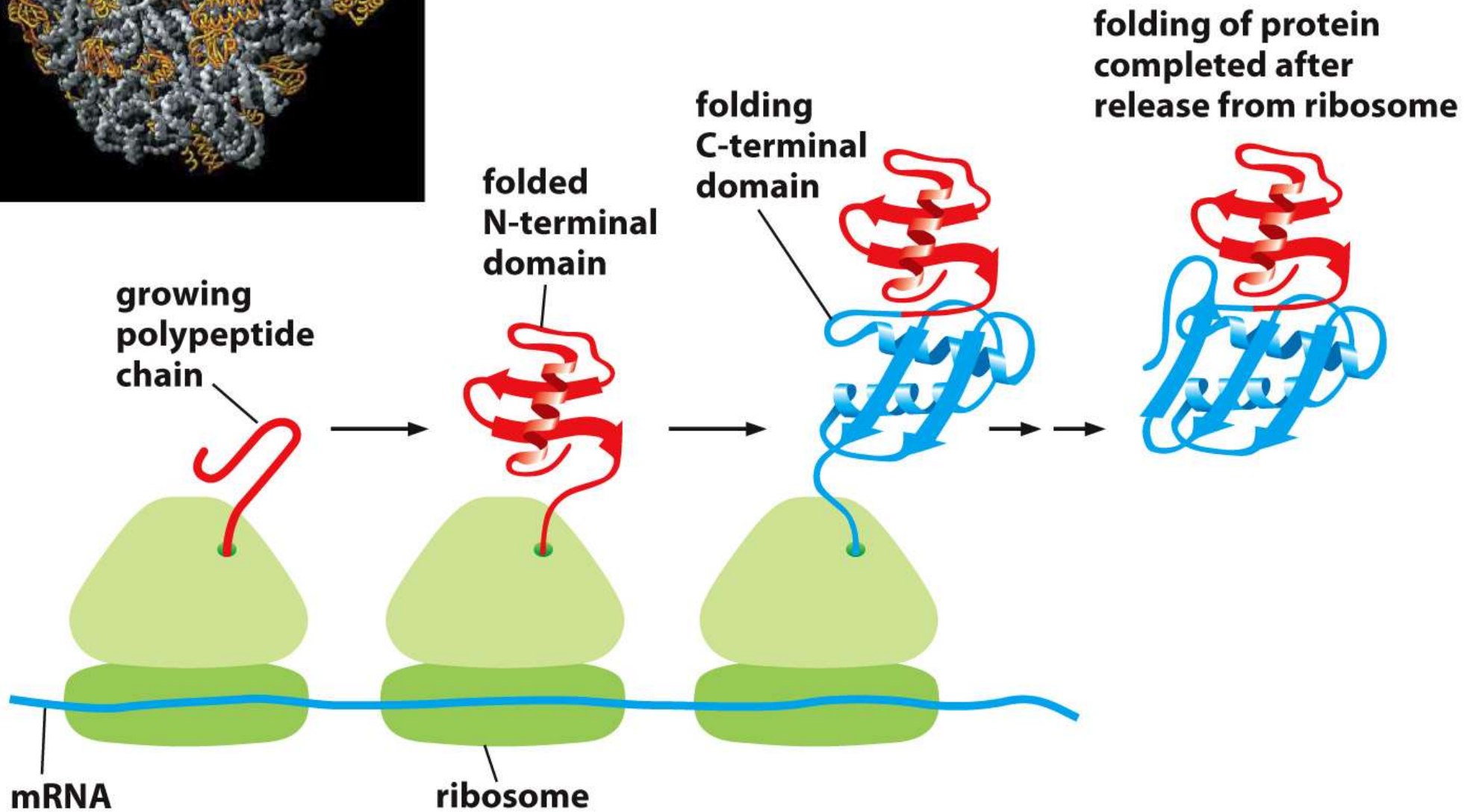
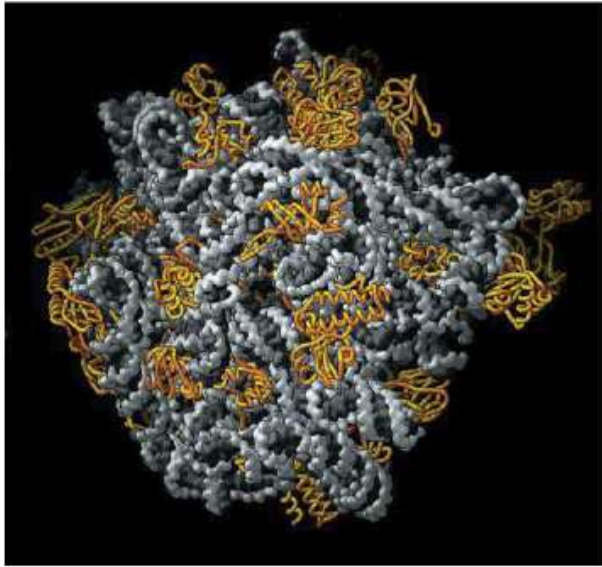
Adapted from J. Lykke-Andersen et al., *Cell* 103:1121–1131, 2000.

Quality control: nonsense-mediated mRNA decay (2)

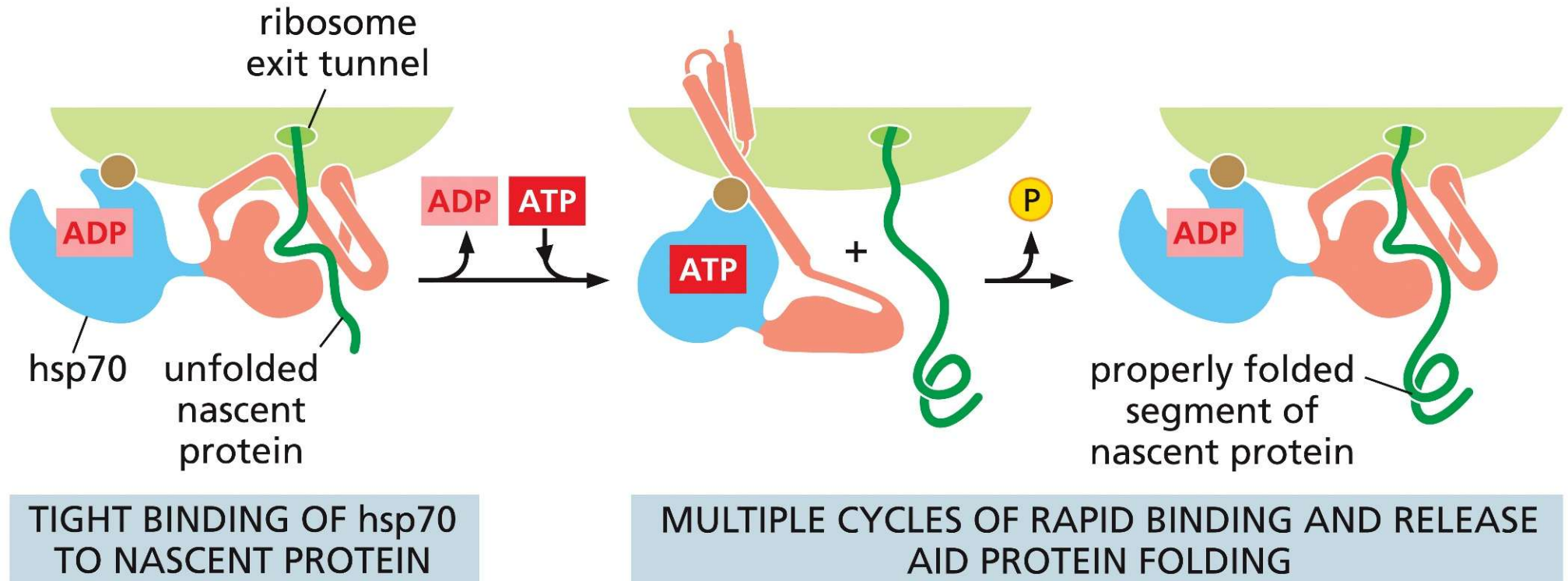


Adapted from J. Lykke-Andersen et al., *Cell* 103:1121–1131, 2000.

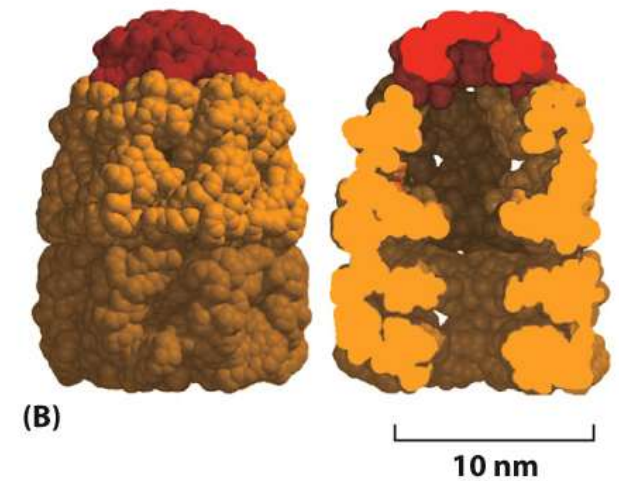
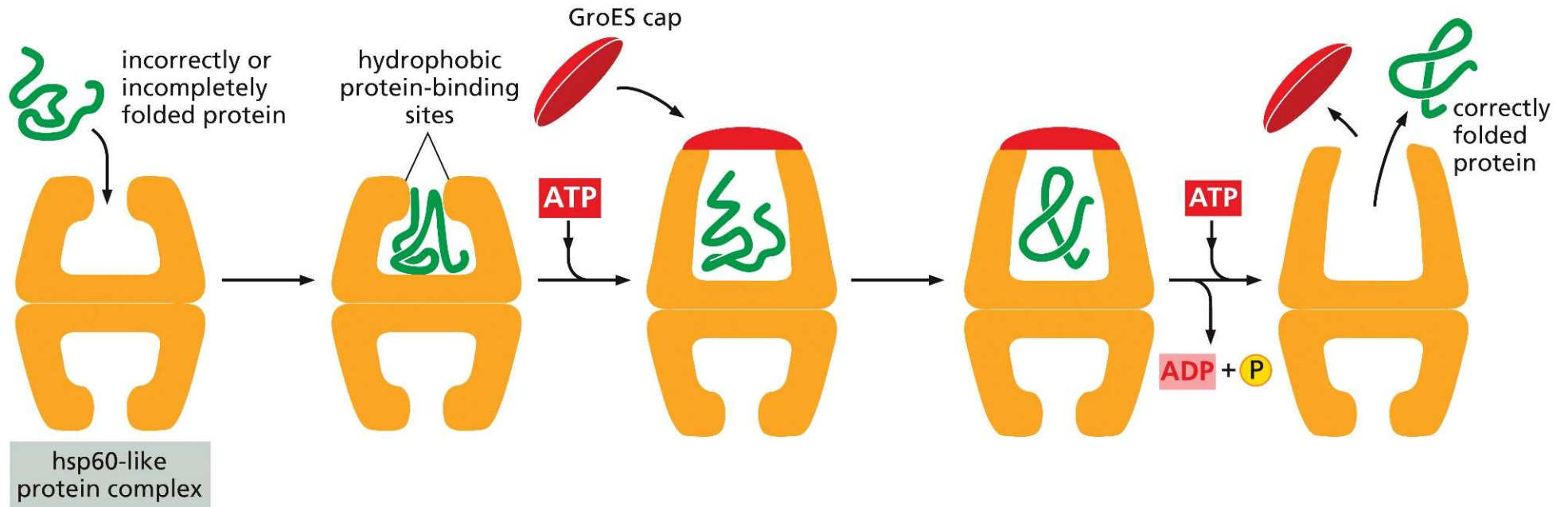
Folding / chaperones

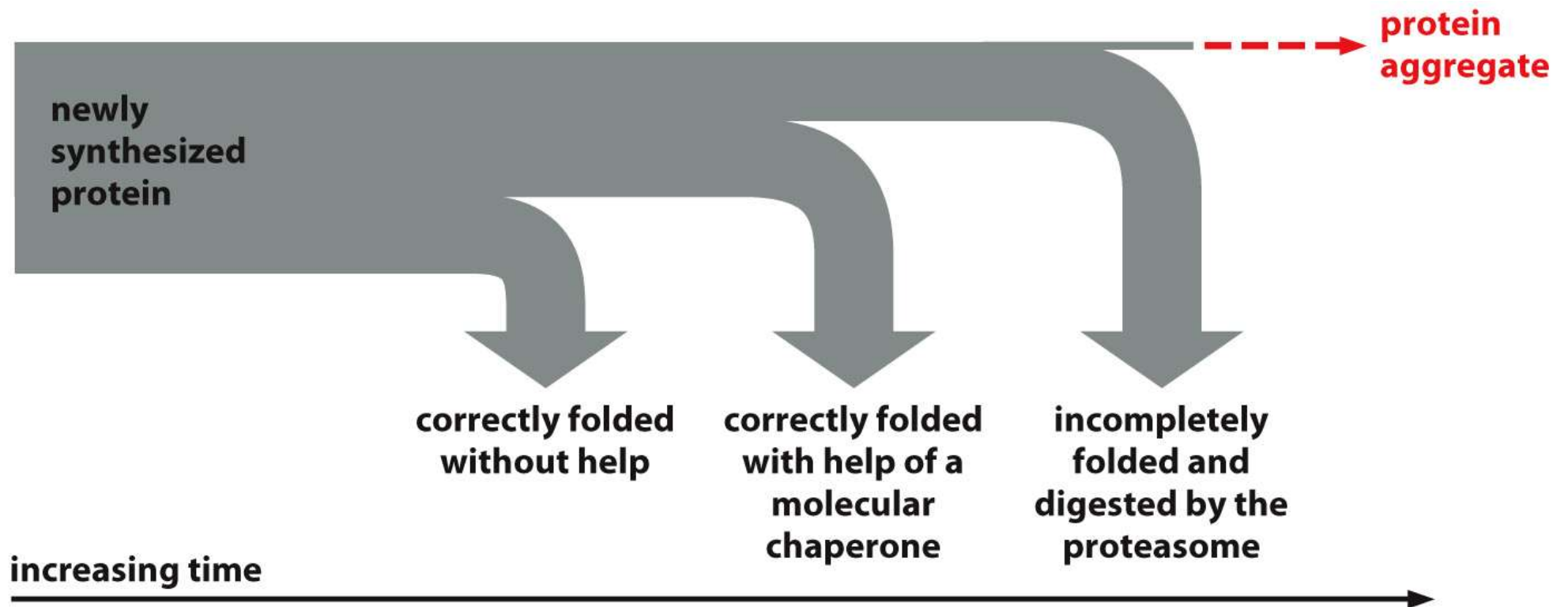


Folding / chaperones; HSP70

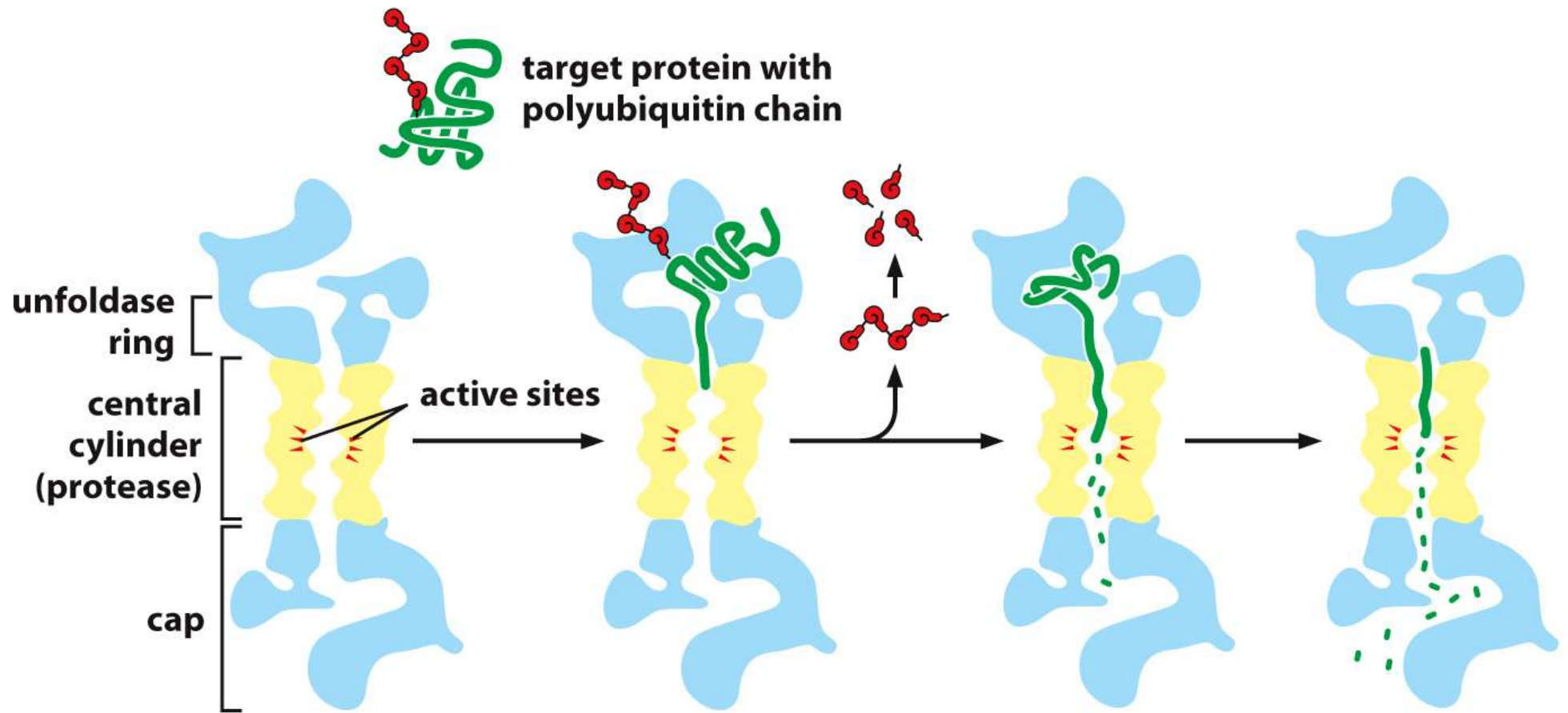


Folding / chaperones; HSP60





The proteasome



Ubiquitin ligase

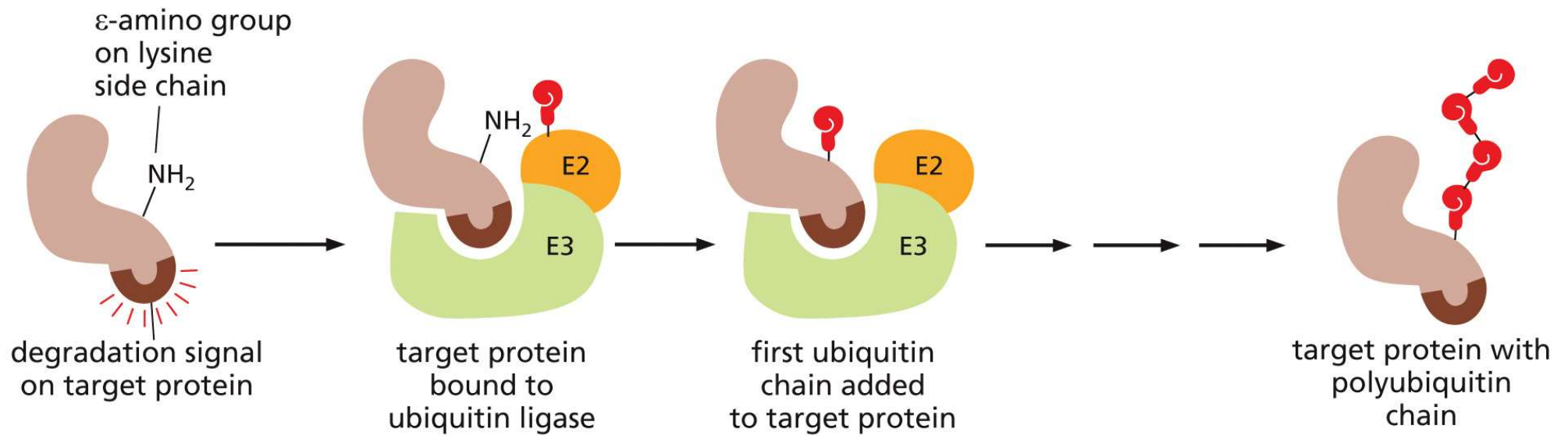
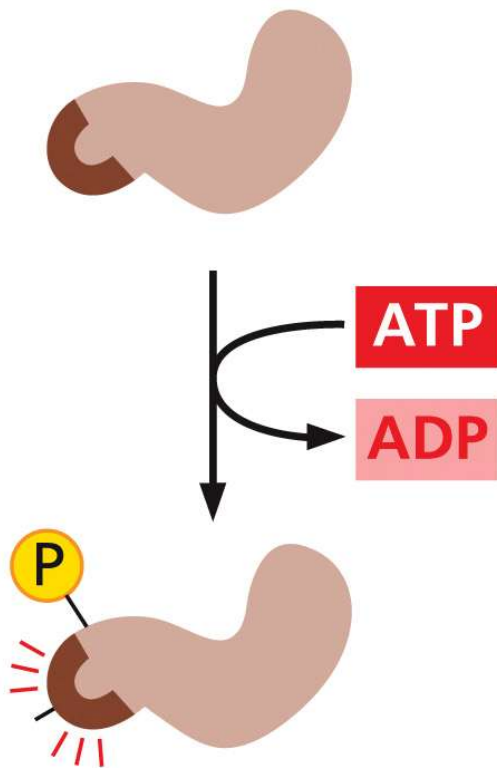


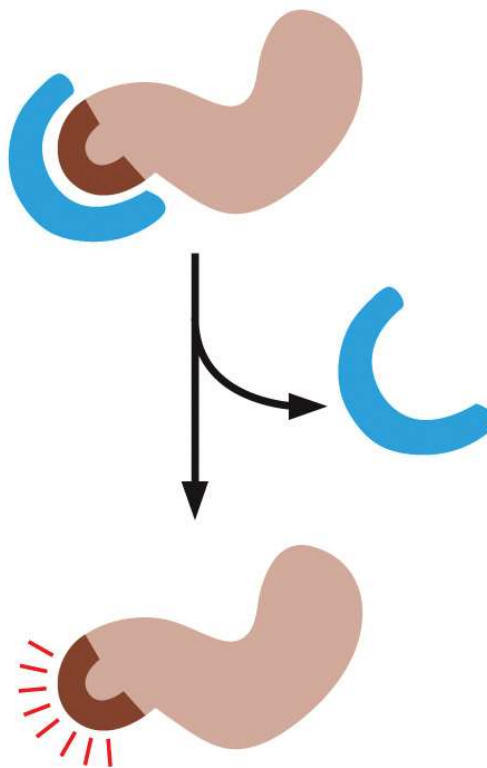
Figure 3-66B

Regulated destruction

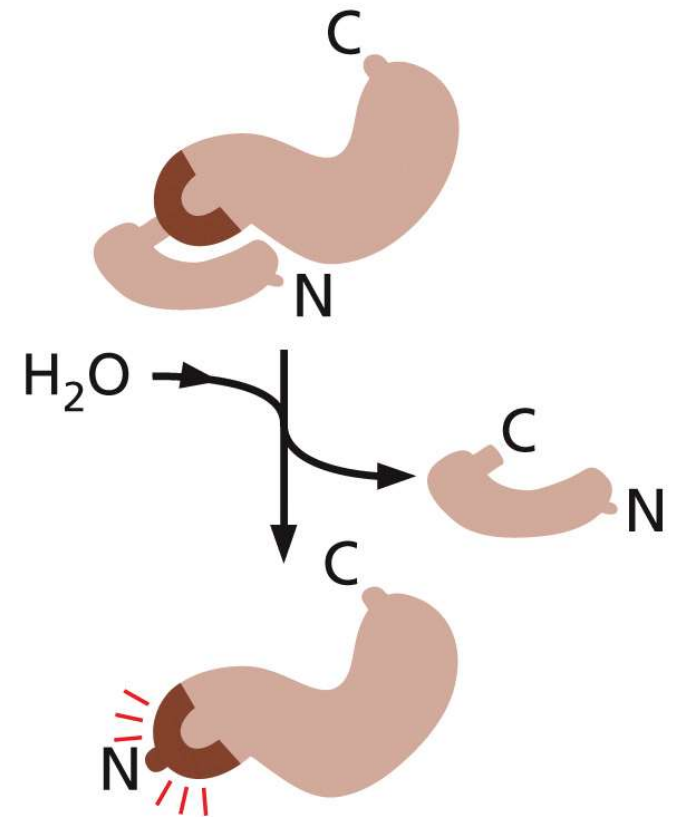
ACTIVATION OF A DEGRADATION SIGNAL



phosphorylation
by protein kinase



unmasking by
protein dissociation



creation of destabilizing
N-terminus

Learning Goals

- mRNA as a particle, nucleolus
- tRNA, tRNA synthetase,
- codon, wobble position
- basics of ribosome
- translation: start / elongation / termination
- puromycin
- mRNA quality control, nonsense mediated decay
- chaperones
- proteasome, ubiquitin system, regulated destruction